

DEDICATED PRELIM PROGRAM(DPP)

UPSC-CSE PRELIMINARY TEST SERIES – 2021/2022

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TEST BOOKLET SERIES

Test Code:

DPP

TEST BOOKLET

GENERAL STUDIES

TEST -07

E7

Time Allowed: Two Hours

Maximum Marks : 200

INSTRUCTIONS

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET DOES NOT HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC.

2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE A, B, C OR D AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.

3. You have to enter your Roll Number on the Test Booklet in the provided alongside this line. DO NOT write anything else on the Test Booklet.

4. This Test Booklet contains 100 questions. Each question is printed in English. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose ONLY ONE response for each item.

5. All items carry equal marks.

6. Penalty for wrong answers:

THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.

- I. There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, One-Third of the marks assigned to that question will be deducted as penalty.
- II. If a candidate gives more than one answer, it will be treated as a wrong answer even if one of the given answers happens to be correct and there will be same penalty as above to that question.
- III. If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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TOPICS COVERD: Solar System, The Origin and Evolution of the Earth Interior of the Earth Distribution of Continents & Oceans Minerals and Rocks Landforms and their Evolution Geomorphic Processes



Q.1) Consider the following statements:

1. The interior of the Earth is constantly shaped by the exogenic forces.
2. The properties of Earth's materials at different depths are governed by the temperature and pressure.

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.2) Consider the following statements regarding the Earthquakes:

1. The point where the energy of the Earthquake is released inside the earth is called the epicentre.
2. The direction of the Earthquake waves depends upon the density of materials it meets.
3. Lesser the density of the material, more is the velocity of the earthquake waves.

Select the incorrect answer using the codes given below:

- a) 1 and 3 only
- b) 2 and 3 only
- c) 1 and 2 only
- d) None of the above

Q.3) Consider the following Earthquake waves.

1. P-waves
2. S-waves
3. Surface waves

What is the correct order of arrival of these waves on the Earth's surface in ascending order?

- a) 3-2-1
- b) 3-1-2
- c) 1-2-3
- d) 1-3-2

Q.4) Which of the following statements is most appropriate regarding the Earthquake waves?

- a) P-waves are responsible for creating density differences in the material they pass through.
- b) The shadow zone of S-wave is smaller as compared to the P-waves.
- c) Body waves are the most damaging of all the earthquake waves.
- d) Most of the earthquake waves vibrate parallel to the direction of wave propagation.

Q.5) Consider the following phenomenon:

1. Construction of large reservoirs
2. Excessive mining activities
3. Gravitational force
4. Volcanic eruptions
5. Strong tidal waves

Which of the above phenomena is/are responsible for the occurrence of the Earthquakes?

- a) 2, 3 and 4 only
- b) 1, 2 and 5 only
- c) 1, 2 and 4 only
- d) 1, 2, 3, 4 and 5

Q.6) Consider the following pairs:

Phenomena	Associated layers
1) Moho Discontinuity:	Mantle- core
2) Gutenberg's Discontinuity:	Crust- mantle
3) Asthenosphere:	Mantle only
4) Lithosphere:	Crust only

Which of the above pairs is/are incorrectly matched?

- a) 1, 2 and 4 only
- b) 1, 2 and 3 only
- c) 2, 3 and 4 only
- d) None of the above

Q.7) Consider the following statements:

1. Shield volcanoes are the most explosive volcanoes made up of basaltic lava.
2. Deccan Trap is an example of a composite volcano largely composed of pyroclastic lava.

Select the correct answer from the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.8) Which of the following consequences would follow if the axis of the Earth were perpendicular to the plane of its elliptical orbit of revolution?

- a) Varying lengths of day and night at different parts of the earth.
- b) Occurrence of equinox at exactly middle of the year.
- c) Equal duration of day and night at different parts of the earth.
- d) Whole year of darkness below the Antarctic circle and vice versa for the Arctic.

Q.9) Consider the following statements regarding the interior of the Earth:

- 1) The continents of the Earth are said to be floating on the oceanic crust.



2) The oceanic crust is made of the granitic rocks **while the continents are formed of basaltic rocks.**

Select the correct answer using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.10) Consider the following statements:

- 1. Igneous rocks are crystalline in structure and may include some traces of fossils.
- 2. Plutonic igneous rocks are the ones that are cooled below the surface of the earth.
- 3. Marble, graphite, and schists are the important examples of igneous rocks.

Select the correct answer using the codes given below:

- a) 1 and 3 only
- b) 2 only
- c) 2 and 3 only
- d) 3 only

Q.11) Consider the following statements:

- 1. Tropics in both the Hemisphere are the maximum extent of the sun's motion.
- 2. Oblique rays during winters are responsible for the lower temperature conditions.
- 3. Periods of dawn and twilight are longer in the tropical latitudes.
- 4. June solstice is characterised by the shortest day and longest nights.

Select the correct answer using the codes given below:

- a) 1 and 3 only
- b) 2 and 3 only
- c) 1 and 2 only
- d) 3 and 4 only

Q.12) Which of the following is the most appropriate regarding seasons on Earth?

- a) Rotation of Earth around its axis is responsible for the occurrence of seasons.
- b) Effect of seasonality is less pronounced in the areas within the tropics.
- c) Axial tilt of the Earth is responsible for the early arrival of spring in the Northern Hemisphere.
- d) There is a positive heat balance during the December solstice in the Northern Hemisphere.

Q.13) With reference to the Plate tectonics, consider the following statements-

1. Convection current in mantle is one of the driving forces behind the movement of the plate.

2. Foci of the earthquake in the mid-ocean ridge area are deep.

3. Continental rise is a part of Mid-oceanic Ridge.

Which of the statements given above is/ are correct?

- a) 2 only
- b) 2 and 3 only
- c) 1 only
- d) All of the above.

Q.14) Consider the following Statements-

- 1. No Volcanic activity is found in the Transform Plate Boundaries.
- 2. Abyssal Plains are extensive plains that lie between the continental margins and mid-oceanic ridges.
- 3. In case of Ocean-Ocean Convergence, the older plate subducts beneath the relatively younger plate.

Select the correct answer using the codes below-

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 2 only

Q.15) In the context of Hotspots, consider the following statements-

- 1. Hotspots develop near the Plate Boundaries.
- 2. Hot spots can be used to track plate movements.

Which of the statements given above is/ are correct?

- a) 1 and 2 only
- b) 1 only
- c) 2 only
- d) Neither 1 nor 2

Q.16) In the context of Distribution of Continents and oceans, consider the following statements-

- 1. The locations of oceans and land masses influences the Earth system.
- 2. The basaltic magma erupts out of the Mid-Oceanic Ridges.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2 only
- d) Neither 1 nor 2

Q.17) With reference to the Placer Deposits, consider the following statements-

- 1. Placer Deposits can be found in the rivers.
- 2. Minerals that form placer deposits are chemically resistant to weathering.



3. Placer Deposits are one of the evidences in the support of the Sea Floor spreading theory.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above.

Q.18) In the context of the Plate Tectonics, consider the following statements-

- 1. The plate boundaries influence the distribution of natural resources.
- 2. Gold deposits are found in the rocks of the active volcanoes.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.19) Consider the following statements about the Continental Drift Theory and Plate Tectonics-

- 1. Plate tectonics explains the reason about why the continents are moving whereas the theory of continental drift did not provide an
- 2. Plate tectonics explain the movement of Continents only whereas the Continental Drift explains the movement of both Continents and Oceans.

Which of the above statements given above is/ are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.20) In the context of the distribution of continents and oceans, consider the following statements-

- 1. The Mid-Atlantic Ridge runs through Iceland.
- 2. Japan is situated near a destructive plate margin.
- 3. The East African Rift Valley is a developing convergent plate boundary.

Which of the statements given above is/ are correct?

- a) 3 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1 and 3 only

Q.21) With respect to Metamorphic Rocks, consider the following statements-

- 1. Metamorphic rocks are formed during plate tectonic processes.
- 2. Metamorphic Rocks are the basis of the Rock Cycle.
- 3. Dynamic metamorphism occurs along the fault zones.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1 and 2 only

Q.22) In the Context of Rocks of India, consider the following statements-

- 1. Dharwar System rocks are metamorphosed rocks.
- 2. Cuddapah System rocks have been formed as a result of erosion and sedimentation of the rocks of Dharwar system.
- 3. Archean rocks are the oldest rocks of the earth's crust.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above.

Q.23) With respect to the Minerals of India, consider the following statements-

- 1. The Jaduguda Mines of Jharkhand is the largest Iron ore Mine of India.
- 2. India has more than 50 per cent of the world's Uranium reserves.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.24) Consider the following Pairs-

Rocks	Types
1. Shale	Sedimentary Rock
2. Quartzite	Igneous Rock
3. Schist	Metamorphic Rock

Which of the pairs given above are correctly matched?

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1 only



Q.25) With reference to different types of rocks, consider the following statements-

1. Foliation is the salient feature of Sedimentary rocks.
2. Sedimentary rocks serve as a record of the events and processes that shaped the surface of Earth.
3. Extrusive Igneous Rocks are characterised by their large grain textures.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) 3 only

Q.26) Consider the following:

1. The mineral composition of the parent rock.
2. Amount of pressure applied.
3. Presence of Water.
4. Temperature Stress.

Which of the above might be the responsible factors that control metamorphic processes?

- a) 1, 2 and 4 only
- b) 2, 3 and 4 only
- c) 2 and 4 only
- d) All of the above.

Q.27) In the context of Minerals and Rocks, consider the following statements-

1. Basalt rocks are created at mid-ocean ridges.
2. Metamorphism may change the mineral composition and the texture of the rock.
3. The most common igneous rock found at the earth's surface is Marble.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) All of the above.

Q.28) With respect to Bauxite, consider the following statements-

1. Bauxite is a sedimentary rock.
2. Telangana is the largest Bauxite producer in India.
3. The Bayer process is a method of refining bauxite to produce alumina.

Which of the statements given above is/are incorrect?

- a) 2 only
- b) 1 and 3 only
- c) 2 and 3 only

d) 3 only

Q.29) Consider the following statements-

1. Sulphur, Aluminum and nitrates are examples of non-metallic minerals.
2. Rajasthan has the largest reserves of copper ore in India.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.30) Consider the following about the Sedimentary Rocks-

1. Sedimentary rocks have layers because of continuous outpouring of lava in layers.
2. These rocks can be a source of major energy resources.
3. Sedimentary rocks tend to be more porous than igneous rocks.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) 2 and 3 only
- d) All of the above

Q.31) In the context of Mica, consider the following statements-

1. Mica is useful as an electric insulator as it does not conduct electricity.
2. No Mica reserves are found in the state of Andhra Pradesh.
3. Mica is widely used in cosmetics and makeup products.

Select the correct answer using the code below-

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) All of the above.

Q.32) With reference to the similarities between the Metamorphic and Igneous Rocks, consider the following statements-

1. Both igneous rock and metamorphic rocks are created under very hot temperatures.
2. Both are formed by the cooling and crystallization of the magma.

Which of the statements given above is/ are correct?

- a) 1 only
- b) 2 only



- c) Both 1 and 2
- d) Neither 1 nor 2

Q.33) Consider the following statements about the Igneous Rocks-

1. Sills are sheet-like igneous intrusions that cut across any layers in the rock they intrude.
2. Stratification occurs only in the Sedimentary Rocks.
3. Dykes are igneous rocks that intrude vertically across the rocks.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 3 only

Q.34) Consider the following processes of erosion and landform development:

1. Attrition
2. Ablation
3. Deflation
4. Abrasion

Which of the above are common to both Aeolian and Fluvial erosion cycles?

- a) 1 and 3 only
- b) 1 and 4 only
- c) 1,2 and 4 only
- d) 2,3 and 4 only

Q.35) In the context of geomorphology, 'Tomboles' and 'Sand Spits' are

- a) Depositional landforms formed by wind action
- b) Erosional landforms formed by action of waves and currents
- c) Erosional landforms formed wind action
- d) Depositional landforms formed by actions of waves and currents

Q.36) "This landform is characterised by dry, vegetation-free, flat bottom depression at the lowest part of a desert basin. It is periodically covered by water that slowly filtrates into the groundwater system or evaporates into the atmosphere."

The landform in the above excerpt is

- a) Hamada
- b) Playa
- c) Zeugen
- d) Yardang

Q.37) Consider the following statements:

1. Some rivers in India are antecedent in nature.
2. All stalagmites found naturally in India are located in Meghalaya.
3. Some portion of the Indian coastline is now submerged.
4. All volcanoes in India are now dormant.

Which of the above statements are correct?

- a) 1 and 3 only
- b) 1,2 and 4 only
- c) 1,3 and 4 only
- d) 2 and 3 only

Q.38) With respect to Coral reefs, consider the following statements:

1. While barrier reefs grow near the coastline around islands and continents, Fringing Reefs are separated from the continent by deeper and wider lagoons.
2. Atolls are usually located in the middle of the sea.
3. Coral reefs are generally absent in the Western margins of continents.

Which of the above statements is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 2 only
- d) 1 and 3 only

Q.39) Where would you find 'Gloups', 'stacks' and 'stumps'?

- a) In the Limestone topography of Andhra Pradesh
- b) Somewhere near the coast of Andaman and Nicobar islands
- c) Towards the border between India and Pakistan across the Thar desert
- d) Near the Kalapani region of the Indo-Nepal border

Q.40) Consider the following statements:

1. Tides refer to the rise and fall of sea levels caused by the combined effects of the gravitational forces exerted the Sun and the Earth only.
2. A rip current is a strong current of water which moves directly towards the shore.
3. The formation of a bay takes place where the coast is made of alternate bands of hard and soft rocks.

Which of the above statements is/are correct?

- a) 1 and 2 only
- b) 2 only
- c) 2 and 3 only
- d) 3 only



Q.41) With reference to various drainage patterns, consider the following statements:

1. In India, a Centripetal drainage pattern is seen in the highlands of Amarkantak.
2. In India, a Radial drainage pattern is seen in the Ladakh highlands.
3. In India, no river flows through a rift valley.

Which of the above statements is/are correct?

- a) 1 only
- b) 2 only
- c) 1 and 3 only
- d) None of the above

Q.42) Which of the following processes signify Diastrophism:

1. Orogenic Forces
2. Epeirogenic Forces
3. Earthquakes
4. Plate Tectonics

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1,3 and 4
- d) 1,2,3 and 4

Q.43) Consider the following statements:

1. Block mountains often result from rifting and faulting.
2. Fold mountains are created where two or more of Earth's tectonic plates are pushed together due to tension induced forces.
3. A horst represents a block pushed upward relative to the blocks on either side by the faulting.

Which of the above statements is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1,2 and 3

Q.44) "While travelling through the desert of Australia's Outbacks, you come across an isolated, flat-topped hill or mountain with steep sides that is smaller in area than a plateau."

The landform you encountered is called

- a) Graben
- b) Gloup
- c) Mesas
- d) Fjord

Q.45) Which of the following can be classified as an intrusive landform?

1. Batholith
2. Sill
3. Dykes
4. Lopolith

Select the correct answer using the code given below.

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1,3 and 4
- d) 1,2,3 and 4

Q.46) With reference to 'valleys', consider the following statements:

1. A gorge is a deep valley with very steep to straight sides.
2. A canyon is wider at its top than at its bottom.
3. Canyons commonly form in horizontal bedded sedimentary rocks and gorges form in hard rocks.

Which of the above statements is/are correct?

- a) 1 and 3 only
- b) 1,2 and 3
- c) 2 and 4 only
- d) All of the above

Q.47) Consider the following statements-

1. Weathering of rocks helps in the enrichment of certain ores.
2. Chemical weathering causes physical damage to rock.

Which of the statements given above are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.48) Consider the following statements-

1. Solifluction happens most in mountainous places where a long dry season is followed by heavy rains.
2. Rock slides happen in mountainous regions where mining is done.
3. Undercutting of steep slopes can trigger Mass Movements.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 2 only



Q.49) With reference to Landslides, consider the following statements-

1. Solifluction is a process where soil is saturated and starts flowing over a slope.
2. Landslides can cause seismic disturbances.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2 only
- d) Neither 1 nor 2

Q.50) Consider the following pairs-

Types of Magma	Characteristics
1. Rhyolitic with high gas content	Explosive eruptions
2. Andesitic	Shield volcanoes are formed by Basaltic Magma
3. Basaltic	Low silica content

Select the correct answer using the code below:

- a) 3 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1 and 3 only

Q.51) Consider the following statements-

1. Karst topography is produced due to the chemical weathering.
2. The action of endogenic forces are uniform throughout the earth.
3. Chemical weathering is the first stage in the production of soils.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 3 only

Q.52) In the context of Geomorphic Processes, consider the following statements-

1. Chemical weathering occurs more slowly in deserts as compared to more humid climates.
2. Orogeny is a part of the exogenic forces.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.53) With respect to erosion, consider the following-

1. Erosion leads to clogging of the waterways and rivers.

2. Erosion can decrease crop yields.
3. Soil erosion helps change the landscape of any geographic area.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 only
- c) 1 and 2 only
- d) All of the above.

Q.54) Consider the following pairs-

Type of Erosion	Features
1. Sheet erosion	Erosion through various narrow and zig-zag channels.
2. Gully Erosion	Deep channels are associated features.
3. Rill Erosion	A complete layer is carried along with water in a larger area.

Select the correct answer using the code below-

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) All of the above.

Q.55) With respect to Weathering, consider the following statements-

1. Salt results in the physical disintegration of rocks.
2. Hydration is an irreversible chemical weathering of rocks.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.56) In the context of Endogenic Forces, consider the following statements-

1. Epeirogeny is a mountain building process whereas Orogeny is continental building process.
2. Diastrophic movements are gradual and very slow in nature.
3. Endogenic forces are usually driven by gravity or atmospheric forces.

Which of the statements given above is/are incorrect?

- a) 1 and 2 only
- b) 2 only
- c) 1 and 3 only
- d) 2 and 3 only

Q.57) Consider the following pairs-

Terms	Features
-------	----------



- | | |
|----------------|------------------------------------|
| 1. Exfoliation | A form of mechanical weathering. |
| 2. Abrasion | is caused by frictional force. |
| 3. Stalagmites | are formed by chemical weathering. |

Which of the pairs given above is/are correctly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 2 only
- d) All of the above.

Q.58) Consider the following statements with respect to Soil formation and weathering-

- 1. Topography is an essential active factor in the Soil formation process.
- 2. Plants play an active role in soil formation through Chemical Weathering.
- 3. Erosion plays a important role in Soil redistribution

Which of the statements given above is /are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 only
- d) All of the above

Q.59) Consider the following pairs-

- | Agent of Erosion | Associated Landform |
|------------------|---------------------|
| 1. Wind | Yardang |
| 2. River | U shaped valleys |
| 3. Glaciers | Alluvial Delta |

Select the correct answer using the code below-

- a) 1 only
- b) 3 only
- c) 1 and 3 only
- d) All of the above.

Q.60) Consider the following statements:

- 1. "Blood Moon" occurs during a total lunar eclipse when Earth lines up between the Moon and the Sun.
- 2. A "Supermoon" occurs when the Moon is at its apogee.
- 3. A "Blue moon" is one extra moon in a season, whereas normally there would be four full moons.
- 4. "Harvest Moon" is the full, bright moon that happens closest to the autumn season.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 and 4
- d) 1, 3 and 4 only

Q.61) Consider the following statements about black holes:

- 1. The only black hole to be photographed till now is called "M87".
- 2. The supermassive blackhole at the centre of the milky way is called "Sagittarius A".
- 3. At the end of its lifetime, the Sun will die off as a black hole.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2, and 3

Q.62) Consider the following statements:

- 1. The Sun's magnetic field spreads throughout the solar system via solar winds.
- 2. Sunspots are relatively warmer than other parts of the sun.
- 3. Solar winds have the capacity to disturb the functioning of satellites and power plants on earth.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) 3 only
- d) 1 and 3 only

Q.63) The term "Accretion Burst Event" is associated with which of the following?

- a) Formation of planets.
- b) Formation of stars.
- c) Formation of earth's crust.
- d) Eruption of a volcano.

Q.64) Consider the following statements:

- 1. An aurora is a colourful display of lights in the sky predominantly seen in lower latitudes.
- 2. It is possible to see an aurora in mid and lower latitudes, although less frequently.
- 3. Due to Coronal Mass Ejection (CME), it is possible to see auroras in more places than they usually occur.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 3 only
- d) 1,2 and 3 only



Q.65) Consider the following statements about abyssal plains:

1. These are flat and smooth because of terrigenous deposits.
2. It is that part of the ocean floor where polymetallic nodules can be found.
3. They are covered with coarse grained sediments.

Which of the statements given above is/are correct?

- a) Only 1
- b) 1 and 2
- c) Only 3
- d) All the above

Q.66) With reference to continental shelves, which of the following statements is/are correct?

1. Shelf breaks are very steep points at which continental shelves end.
2. Continental shelves are ecologically diverse and rich feeding grounds.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.67) With reference to the Mid oceanic ridges Consider the following statements:

1. These ridges form the largest mountain system on earth.
2. They are found at the site of convergent boundaries.
3. They are of tectonic origin and provide evidence in support of plate tectonics theory.
4. The longest mid-oceanic ridge is in the Pacific Ocean.

Which of the statements given above is/are not correct?

- a) 1 and 2 only
- b) 1 and 3 only
- c) 2 and 4 only
- d) None of the above

Q.68) Consider the following pairs:

Minor ocean reliefs	Description
1. Mid-oceanic ridges	- Two chains of mountains separated by a large depression.
2. Seamount	- A flat topped mountain submerged in the ocean.
3. Guyots	- Mountain that is volcanic in origin, rising from the sea floor.

4. Submarine canyon - Deep oceanic valleys extending from the mouths of large rivers.

Which of the above pairs is/are incorrectly matched?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 and 4 only
- d) 1 and 4 only

Q.69) With reference to submarine trenches consider the following statements:

1. Submarine trenches are not always located in the middle of the ocean.
2. They can be found running parallel to the continents.
3. The Indian Ocean has the largest number of such trenches.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) 1 and 2 only
- d) 1 and 3 only

Q.70) With reference to "pelagic deposits" found on the ocean floor, which of the following statements is/are correct?

1. They are sediments which are formed from the remains of living beings only and are found on the abyssal plains.
2. "Red clay" is a type of pelagic sediment which is of volcanic origin.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.71) Consider the following regarding Cloud bursts?

1. There is no possibility of them occurring in the plain regions.
2. Such clouds are generally cumulonimbus and may undergo orographic lift.
3. The Copenhagen Climate adaptation model contains ways to manage this event.
4. It involves downward movement of air currents coupled with the presence of frontal cyclones.

Which of the statements given above is/are correct?

- a) 1, 2, and 4 only
- b) 2 and 3 only
- c) 1, 3, and 4 only
- d) 1 and 2 only



Q.72) Consider the following statements:

1. It is not possible to transmit a zoonotic disease through a fungus.
2. The E-2025 initiative deals with the elimination of tuberculosis.
3. Sickle cell anemia is a hereditary orphan disease.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 3 only
- c) 2 and 3 only
- d) 1, 2, and 3

Q.73) Consider the followings statements regarding the Pradhan Mantri Garib Kalyan Ann Yojana:

1. It aims at providing free foodgrains to those poor and migrants who have not been covered under the National Food Security Act.
2. The cost for the supply of grains under PMGKAY is borne by the Centre and State in the ratio of 50:50 for the normal category states.
3. The scheme was introduced as a part of Atma Nirbhar Bharat and is touted as the biggest food security programme of the world.

Which of the statements given above is/are incorrect?

- a) 1 and 3 only
- b) 2 only
- c) 1 and 2 only
- d) 1, 2, and 3

Q.74) In the context of the India H₂ alliance, consider the following statements:

1. It is a government led initiative that deals with the building supply chain in the hydrogen economy.
2. It will aid in the development of National Hydrogen Policy and Roadmap by 2030.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.75) In the context of the Schedules of the Constitution, consider the following statements:

1. The Constitution provides a set of criteria for a language to be listed under the 8th schedule.
2. Articles 29 and 30 of the Constitution deals with the provision relating to Linguistic minorities only.

3. Any change in the 8th Schedule requires an amendment to the Constitution with ratification by half of the states.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 3 only
- c) 1 only
- d) None of the above

Q.76) Consider the following pairs:

Location	Country
1. Nancowry Island group	India
2. La Soufriere Volcano	Laos
3. Plain of Jars	Jamaica

Which of the pairs given above is/are correctly matched?

- a) 1 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) 2 and 3 only

Q.77) Consider the following statements regarding the WTO:

1. The agriculture subsidy under WTO is calculated as a percentage of total volume of a crop produced.
2. There is no upper limit to the subsidies that can be provided under the Green or Blue Box.
3. The Peace clause of the WTO deals with the subsidies under the Agreement on Agriculture

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2, and 3

Q.78) Consider the following statements:

1. Kessler Syndrome occurs when the excessive gravity prevents the conversion of the neutron star to black hole.
2. Red star is the first stage in the expansion of stars and formation of Black hole.

Which of the statements given above is/are incorrect?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.79) In the context of Ramanujacharya, consider the following statements:



1. He propounded the theory of Vashistadavaita dealing with dualism.
2. He belonged to the Sarguna school of thought and believed in atman and brahman.
3. He authored exposition on the Bhagwat Gita and Brahma sutra.

Which of the statements given above is/are correct?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 3 only
- d) 1 and 3 only

Q.80) In the context of the Medical Oxygen, which of the statements given below is/are incorrect?

1. The Cryogenic Distillation process uses the atmospheric air to produce Liquid Medical Oxygen (LMO).
2. Oxygen concentrators are internally equipped with cryogenic tankers to store LMO.
3. Most Oxygen concentrators use Zeolite as the nitrogen absorbing materials.
4. Medical oxygens stored in an Oxygen cylinder are in liquid form.

Select the correct answer using the codes given below:

- a) 1 and 4 only
- b) 1, 3, and 4 only
- c) 2 and 4 only
- d) 2 and 3 only

Q.81) With reference to the UN Food Systems Summit 2021, consider the following statements.

1. The UN Secretary General called for the second session of the UN Food Systems Summit to be held in the General Assembly in September 2021.
2. Objective is to analyze the roadmap to realize the vision of the 2030 Agenda for Sustainable Development.
3. Process involved in producing, processing, transporting and consuming food is termed as food systems.
4. The Ministry of Food Processing and Industries has created a website to seek input from various stakeholders in context with the UN Food System Summit-2021.

Which of the above given statements is/are correct?

- a) 1, 2 and 3 only
- b) 1 only
- c) 2 and 3 only
- d) All of the above

Q.82) With reference to the recently launched Supply Chain Resilience Initiative, consider the following statements.

1. The quad group of countries launched it to increase trade in the Indo-Pacific region.
2. Supply Chain Resilience is a concept in which a country is not dependent on any one country for supply chains but many countries.
3. It aims to create a virtuous cycle of enhancing supply chain resilience with a view to eventually attaining strong sustainable, balanced and inclusive growth in the Indo-Pacific region.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 1 only
- c) 2 and 3 only
- d) All of the above

Q.83) Which of the following statements is correct with respect to Convalescent Plasma Therapy?

1. It is based on the age old concept of passive immunity, where there is use of antibodies developed in the recovered patient.
2. The whole blood or plasma from an already infected and cured patient is injected into critically ill patients in this therapy.
3. Plasma is the clear, straw-colored liquid portion of blood that remains after red blood cells, white blood cells, platelets and other cellular components are removed.
4. It was used in the early part of the 20th century to treat patients with Spanish Flu.

Select the correct answer from the codes given below.

- a) 1 and 3 only
- b) 2 and 3 only
- c) 2 only
- d) All of the above

Q.84) With reference to the recent outbreak of suspected African swine fever, consider the following statements.

1. It is a variant of swine flu and has severe human health impacts.
2. It has no cure other than culling of the animals to prevent its spread.

Which of the statements given above is/are true?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.85) With reference to the recently released Global Gender Gap report 2021, consider the following statements.

1. India has fallen places in the overall ranking from its previous report.
2. On account of parity between income of men and women, India lies in the bottom 10 globally on this indicator.
3. Top 10 Gender-EQ.ual countries are all from Europe barring New Zealand.
4. Released for the first time in 2006 by the World Economic Forum.

Select the correct answer from the codes given below.

- a) 1, 2 and 4 only
- b) 3 and 4 only
- c) 1 and 2 only
- d) All of the above

Q.86) With reference to Impacting Research, Innovation and Technology (IMPRINT) India programme, consider the following statements:

1. Initiative of the Ministry of Science and Technology.
2. Joint venture of the Indian Institutes of Technology and the Indian Institutes of Science.
3. It seeks to develop a road for research to solve major engineering and technology challenges in environment and climate along with other domains.

Select the correct answer from the codes given below:

- a) 1 and 2 only
- b) 1 only
- c) 2 and 3 only
- d) All of the above

Q.87) Which of the following is true regarding the proposed amendments to the Forest (Conservation) Act, 1980 (FCA)?

- a) Empowering the state government to lease forest land to private individuals and corporations.
- b) Creation of 'no-go' areas, within the forest, where specific projects would not be allowed.
- c) Proposal to grant exemptions to railways, roads, tree plantations, oil exploration, wildlife tourism and strategic projects in the forests.
- d) All of the above.

Q.88) With reference to the eligibility of the Stand-Up India Scheme, consider the following statements.

1. The borrower under this scheme should be Scheduled Tribes, Scheduled Castes and women entrepreneurs above 18 years of age.

2. Loans under this scheme are available only for green field projects.
3. Borrower should not be a defaulter to any bank, financial institution etc.
4. In case of an enterprise or joint venture, at least 51 % of the shareholding and controlling stake should be held by either a Scheduled Tribes/Scheduled Castes or Women entrepreneur.

Which of the above given statements regarding eligibility is/are true?

- a) 1 and 2 only
- b) 1, 2 and 4 only
- c) 2 only
- d) All of the above.

Q.89) Handheld 'Surface Enhanced Raman Spectroscopy (SERS) - based platform, recently in the news, is used for which of the following.

- a) Used to detect the presence of any foreign objects in the body.
- b) Measure the air Quality by analysing the presence of various air pollutants.
- c) Early diagnosis of dengue.
- d) To map the underground oil reserves of the country.

Q.90) The term 'Gigaton Volume Detector' was recently news and is related to

- a) the study of high energy neutrinos that come from the earth's core.
- b) the study of the interior of the earth on the basis of mass and volume.
- c) the mapping of the estimated huge underground petroleum reserves in the arctic regions.
- d) demarcate and map the Aquifers present on various places and determine the availability of freshwater.

Q.91) With reference to Right to Repair, consider the following statements:

1. ASEAN (Association of SouthEast Asian Nation) has brought a law to establish Right to Repair to promote Circular Economy.
2. This right ensures that spare parts have to be available for upto a decade.

Which of the following statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 & 2
- d) Neither 1 nor 2



Q.92) With reference to the recent change in protection status of a few species of Elephants by IUCN, consider the following statements

1. Savannah Elephant and Asian Elephant now have the same Protection status of Endangered
2. The Asian Elephant, which is found in India are smaller than the African Elephants
3. Elephant has the longest gestation period among all mammals

Which of the above statements is/are correct?

- a) 1 & 2 Only
- b) 2 & 3 only
- c) 1 & 3 only
- d) 1, 2 & 3

Q.93) With reference to Local reservation in private sector (for example: Haryana passed recently Haryana State Employment of Local Candidates Act, 2020), consider the following statements:

1. Haryana becomes third state after Andhra Pradesh and Tamilnadu, to have 75% reservation for its residents on the basis of domicile of the state.
2. With reference to reservation, it was Indra Sawhney Case 1992 Judgement, which stated that Reservation shouldn't cross 50%.
3. Article 14, 15, 16 and 19 guarantee freedom of movement and consequent employment in India.

Which of the above statement is/are correct?

- a) 1 & 2 only
- b) 2 & 3 only
- c) 1 only
- d) 1, 2 & 3

Q.94) Consider the following statements with reference to Borneo Island, which was recently in news for discovery of a songbird species:

1. Borneo Island is governed by two separate sovereign nations
2. Borneo Island is the largest island of Asia
3. Borneo Island is located to the West of Sumatra

Which of the above statements is/are correct?

- a) 1 & 2 only
- b) 2 only
- c) 3 only
- d) 2 & 3 only

Q.95) With reference to Sugamya Bharat App, launched under Sugamya Bharat Abhiyan, by Ministry

of Social Justice and Empowerment, Consider the following statements:

1. Aadhar is not mandatory to sign in to Sugamya Bharat App
2. Sugamya Bharat App will enable people with disability as well as elderly, to register complain related to accessibility problems in buildings, in modes of transport or any infrastructure
3. Sugamya Bharat Abhiyan initiative is in line with United Nations Convention on Rights of person with Disability

Which of the following statements is/are correct?

- a) 1 & 2 only
- b) 3 only
- c) 1 & 3 only
- d) 1, 2 & 3

Q.96) With reference to Critical Information Infrastructure (CIIs) in India, consider the following statements:

1. Critical Information Infrastructure is not defined in any legislation in India
2. India has established National Critical Information Infrastructure Protection Centre (NCIIPC) to regulate and protect critical Information Infrastructure

Select the correct Statement using the codes given below:

- a) 1 only
- b) 2 only
- c) Both 1 & 2
- d) Neither 1 nor 2

Q.97) With reference to the Report - 'State of the world's indigenous peoples: Rights to Land, territories and resources', Consider the following statements:

1. United Nations Department of Economic and social affairs released this report
2. Indigenous peoples are inheritors and practitioners of unique cultures and ways of relating to people and the environment
3. Indigenous and Tribal Peoples Convention 1957 was the first convention to deal specifically with Indigenous rights in independent countries

Which of the above Statements is/are correct?

- a) 1 & 2 only
- b) 2 & 3 only
- c) 1 only
- d) 1, 2 & 3

Q.98) Consider the following statements:



1. Heavy metals are metals with relatively high density and toxic at low concentrations
2. Blue baby syndrome and Black foot disease are caused by Nitrates

Which of the following statements is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 & 2
- d) Neither 1 nor 2

Q.99) Hypnea Indica and Hypnea Bullata are species of

- a) Insectivorous Plants
- b) Sea weeds
- c) Invasive Flora
- d) Epiphytes

Q.100) With reference to Traditional Indian toys and their respective states, consider the following pairs :

	Indian Toys	Staes
1	Channapatna	Bihar
2	Choppu Saman	Tamilnadu
3	Ghuggu	Punjab
4	Nirmal Toys	Telangana
5	Thanjavur Thalaiyatti Bommai	Kerala

Which of the above pairs is correctly matched?

- a) 1, 3 & 5 only
- b) 2, 3 & 4 only
- c) 2, 4 & 5 only
- d) 1, 3 & 4 only



Q.1) Ans: B

Exp:

- **Statement 1 is incorrect:** It is not the interior of the earth rather the surface of the earth that is constantly shaped by the exogenic forces. **Endogenic forces** are the ones that act from the interior of the earth like Volcanism, Diastrophism, etc. They are responsible for the building or elevation of the reliefs on Earth. **Exogenic processes** on the other hand are the land wearing processes. Weathering, landslides, erosion, and deposition are some of the examples of these. They are external processes that act on the surface of the Earth.
- **Statement 2 is correct:** Understanding about the variation of temperature and pressure from the crust to the core helps in estimating the materials that form the interior of the Earth. The density of the materials also varies from the surface towards the core of the earth. Knowing the temperature, pressure and density at different depths, it is possible to estimate the materials that are distributed in the interior of the Earth.

Q.2) Ans: A

Exp:

- **Statement 1 is incorrect:** The point where the energy of the Earthquake is released inside the surface of the Earth is called the **focus** or the **hypocentre**. Whereas the point on the surface of the earth nearest to the focus is called the epicentre of the Earth. **Epicentre** is the point on the surface of the Earth that is the first to experience the earthquake waves.
- **Statement 2 is correct:** The energy is released by the Earthquakes in the form of waves. There are two kinds of waves: **Body waves** and the **Surface waves**. Body waves are the ones that travel along the body of the surface of the Earth. They are first ones to be released from the focus whereas Surface waves are generated when the body waves encounter the surface of the Earth. The direction of the wave changes as it undergoes reflection or

refraction when in contact with materials of different densities.

- **Statement 3 is incorrect:** Denser the material, greater is the velocity of the wave and vice versa. Hence, the velocity of the wave also depends on the density of the material it encounters.

Q.3) Ans: C

Exp:

- **Statement C is correct:** P-waves also called as the Primary wave is the first one to report on the surface of the Earth. They can pass through all solid, liquid and gas and are similar to sound waves. S-waves on the other hand arrive after a certain time delay. They can only travel through solids. Surface waves are the last to arrive at the surface. They are the most destructive of the other two.

Q.4) Ans: A

Exp:

- **Statement 1 is correct:** The vibrations of P-waves are parallel to the direction of motion of the earthquake waves. They cause stretching and squeezing of material they pass through. Hence, they are responsible for the creation of density difference in the material they pass through. The vibrations of all other waves are perpendicular to the direction of the propagation of the wave.
- **Statement 2 is incorrect:** S-waves can only pass through solids while P-waves can pass through solid, liquids, and gases. Both P-waves and S-waves are recorded up to 105 degrees from the location of the seismograph while beyond that S-waves are not recorded. P-waves are also recorded beyond 145 degrees. Hence, the shadow zone of P-waves is in the form of band between 105 to 145 degrees while that of S-waves is all area beyond 105 degrees. Thus, the shadow zone of S-waves is much larger than the P-waves.
- **Statement 3 is incorrect:** Surface waves are the most damaging waves in terms of losses caused to the life and property of the people.



They are the last to strike the surface of the earth and are the ones actually responsible for the collapse of the structures.

- **Statement 4 is incorrect:** Most of the Earthquake waves vibrate in the direction perpendicular to the direction of the propagation of the waves. It is only the P-wave that moves in the direction parallel to the propagation of the wave thereby creating a density difference in the material it passes through.

Q.5) Ans: C

Exp:

- **Statement 1 is correct:** It is quite possible that the earthquake will occur in the areas of large reservoirs as huge structures cause pressure on the underlying rocks. Such earthquakes are called **reservoir induced earthquakes**.
- **Statement 2 is correct:** Collapse of the roofs of the underground mining structures can also cause earthquakes. Such earthquakes are referred to as the **collapse earthquakes**.
- **Statement 3 is incorrect:** Earthquakes are the sudden movement of the rocks below the surface of the earth along a fault line. The rocks under the Earth's surface are held together by the force of friction between them. The pressure of the overlying rock strata exists over them and at some point their tendency to slide past each other exceeds the force of friction causing their movement and ground shaking. Such earthquakes are called the **tectonic earthquakes** and the most commonly caused natural earthquakes. The presence of gravitational force itself does not produce any earthquake.
- **Statement 4 is correct:** In the areas of active volcanoes, there exists **volcanic earthquakes**. They are also a kind of tectonic earthquakes.
- **Statement 5 is incorrect:** Strong tidal waves are itself not responsible for the occurrence of earthquakes. However, in cases where the epicentre of the earthquake lies under a body of water, there occur strong tidal waves in the form of Tsunamis.

Q.6) Ans: A

Exp:

- **Statement 1 is incorrect:** Moho's discontinuity exists between the mantle and crust interface. Discontinuity is an interface where the seismic waves show change in their velocity.
- **Statement 2 is incorrect:** Gutenberg's discontinuity exists between the mantle and the core interface.
- **Statement 3 is correct:** Asthenosphere exists in the upper part of the mantle. This layer houses the magma that finds its way out during a volcanic eruption. It is a weak viscous layer extending upto 400 km.
- **Statement 4 is incorrect:** Lithosphere is the layer comprising the core and upper most solid part of the mantle. It lies between 10-200 km from the surface of the earth.

Q.7) Ans: D

Exp:

- **Statement 1 is incorrect:** Shield volcanoes are the largest of all volcanoes as they are mostly composed of basalt. Basalt is a highly fluid lava that prevents these volcanoes from developing steep structures. They explode rarely and in situations like the entering of water into their vents. Hence, they are made of basaltic lava but are not the most explosive ones. Calderas are regarded as the most explosive volcanoes. Their explosion is so strong that they tend to collapse on themselves.
- **Statement 2 is incorrect:** Deccan Traps are an example of flood basalt province and not composite volcano. Large numbers of pyroclastic materials and ashes are usually erupted by the composite volcanoes which are highly violent and erupt lava that is cooler and more viscous than basalt. Basalt Flood province is created when highly fluid basaltic lava is erupted and travels to a large distance in the surrounding area forming a thick sheet of lava. **Deccan traps** in Maharashtra are an example of flood basalt province.



Q.8) Ans: C

Exp:

- **Statement 3 is correct:** The axis of Earth is inclined at an angle of 66 degrees from the plane of its elliptical orbit of revolution. The result of this tilt of earth's axis is responsible for varying lengths of day and night at different locations on the Earth and for giving rise to different seasons. If this axis was inclined perpendicular to its orbit, then all places on earth would experience the same lengths of day and night. Hence, the duration of day and night at all places on the earth would be the same.

Q.9) Ans: A

Exp:

- **Statement 1 is correct:** The outermost layer of the Earth or the crust is of two types- Continental crust and the Oceanic crust. Continental crust is made of the granitic rocks and mainly composed of Silica and Aluminum. Hence, it is also known as the Sial. It is less dense than the Oceanic crust. Oceanic crust is made up of the denser basaltic rocks and its main constituents include the silica, iron and magnesium. Hence, it is also referred to as Sima. Since, the oceanic crust is denser than the continental crust, the continents are regarded as floating on the oceans.
- **Statement 2 is incorrect:** The oceanic crust is made of denser basaltic rocks and the continental crust is made of the granitic rocks. The density of the oceanic crust is more than that of the continental crust. Hence, it is found below the continental crust.

Q.10) Ans: B

Exp:

- **Statement 1 is incorrect:** Igneous rocks are crystalline in structure and are devoid of any fossils or any layers. They are formed by the

cooling and solidification of magma under the Earth's crust.

- **Statement 2 is correct:** Igneous rocks are either **Plutonic** or **Volcanic rocks**. Plutonic rocks are formed when magma cools below the surface of the Earth while Volcanic rocks are formed when the magma cools on the surface of the Earth after its way out of a volcanic eruption.
- **Statement 3 is incorrect:** Marble, graphite, schists are the examples of the metamorphic rock and not igneous rocks. Examples of igneous rocks include basalt, granite, etc.

Q.11) Ans: C

Exp:

- **Statement 1 is correct:** The movement of the sun is restricted between the Tropic of cancer in the Northern Hemisphere and the Tropic of Capricorn in the Southern Hemisphere.
- **Statement 2 is correct:** During winters in the Northern Hemisphere, the sun is overhead at the Tropic of Capricorn. Hence, direct rays fall on the Southern hemisphere while oblique rays travel to the Northern Hemisphere. These rays travel a longer distance and hence lose much of their heat in their path in the atmosphere. Thus, during winters temperature is lower and the sun rays fall over a larger surface of the earth in the Northern Hemisphere.
- **Statement 3 is incorrect:** Twilight is the duration between the onset of sunset and complete darkness. Dawn refers to the duration between the onset of sunrise and full daylight. The sun is below the horizon during the dawn or twilight and Earth receives diffused or refracted light. Since, oblique rays fall on the temperate latitudes, the duration of dawn or twilight is longer whereas near the tropics, this duration is almost negligible.
- **Statement 4 is incorrect:** June solstice or the summer solstice (21st June) occurs when the sun is directly overhead at the Tropic of cancer in the Northern Hemisphere. The Northern hemisphere experiences summer and the days are longer while the nights are shorter.



Q.12) Ans: B

Exp:

- **Statement 1 is incorrect:** Rotation of Earth around its own axis is responsible for the occurrence of day and night. Earth takes 24 hours to rotate around its own axis. The part of the earth that faces the sun experiences the day while the other part faces the night.
- **Statement 2 is correct:** Effect of seasonality is less pronounced at the areas within the tropics. Days and nights are almost equal all the year round near the equator as the variation in the sun's position is almost negligible during the day time. Seasons are more distinctly marked in the higher latitudes.
- **Statement 3 is incorrect:** Axial tilt of the earth is responsible for the variation in the length of day and night. Axial tilt along with the revolution of the earth causes the occurrence of seasons with variations in the length of day and night.
- **Statement 4 is incorrect:** December solstice or the winter solstice (December 22) occurs when the sun is overhead at the Tropic of capricorn in the southern hemisphere. The Northern Hemisphere experiences winters and is characterised by lower temperature and negative heat gain. The net gain in the heat occurs in the Northern Hemisphere during the summer season

Q.13) Ans: D

Exp:

- **Statement 1 is correct:** The forces that drive Plate Tectonics include: Convection in the Mantle-Heat within the earth comes from two main sources: radioactive decay and residual heat, Ridge push - gravitational force at the spreading ridges and Slab pull -gravitational force in subduction zones.
- **Statement 2 is correct:** Shallow-focus earthquakes-usually less than 30 km deep- appear to be associated with mid-ocean ridges.Shallow focus earthquakes are usually of large spread, causing greater damage at the surface or the earth's crust.

- **Statement 3 is correct:** The continental margin is the shallow water area found in proximity to continents. The continental margin consists of three different features: the continental rise, the continental slope, and the continental shelf. Continental margins constitute about 28% of the oceanic area.

Q.14) Ans: D

Exp:

- **Pair 1 is incorrect:** Major Volcanoes do not typically occur at transform boundaries.The grinding action between the plates at a transform plate boundary results in rare shallow earthquakes, large lateral displacement of rock, and a broad zone of crustal deformation.
- **Pair 2 is correct:** The Continent-Continent convergence is when two continental plates collide.When two plates of continental crust collide, the material pushes upward. This forms a high mountain range. Example: Formation of the Fold Mountains.
- **Pair 3 is incorrect:** In Ocean-Ocean convergence, two oceanic plates converge or collide. A denser oceanic plate subducts below a less dense oceanic plate forming a trench along the boundary.Constant volcanism above the subduction zone creates layers of rocks. Subduction of an ocean plate beneath another oceanic crust results in a volcanic island arc, an ocean trench, and many earthquakes.

Q.15) Ans: C

Exp:

- **Statement 1 is incorrect:** Hot spot is independent of plate boundaries. Instead it occurs at mantle plumes.A mantle plume is an area under the crust of Earth, where magma is hotter than surrounding magma.Heat from this extra hot magma causes melting and thinning of the rocky crust, which leads to widespread volcanic activity on Earth's surface above the plume.



- **Statement 2 is correct:** The hot spot itself never changes position, but the tectonic plates are constantly moving, so the volcano formed will "move" along with the tectonic plate to the direction where ever the tectonic plate is heading, but at the same time the hot spot doesn't stop producing lava. This enables tracking of the movement of the Plates.

Q.16) Ans: C

Exp:

- **Statement 1 is correct:** How sunlight is absorbed or reflected by the surface of the Earth, Regional temperatures, extent of ice sheets, climate of the area/region etc are influenced by the distribution of the continents and oceans.
- **Statement 2 is correct:** The dominant type of lava resulting from magmatic activity at mid-ocean ridges is basalt, also called mid-ocean ridge basalt (MORB). Basaltic magmas that form the oceanic crust of Earth are generated in the asthenosphere at a depth of about 70 kilometres. MORB is by far the most common rock type on the Earth's surface, as the entire ocean floor consists of it.

Q.17) Ans: A

Exp:

- **Statement 1 is correct:** A placer deposit or placer is an accumulation of valuable minerals formed by gravity separation from a specific source rock during sedimentary processes. Placers can be found in rivers -alluvial placers and on the coast, particularly in beaches -beach placers.
- **Statement 2 is correct:** Placer Minerals must be both dense and resistant to weathering processes. To accumulate in placers, mineral particles must have a high specific gravity. They should also be durable. Example- gold, platinum, etc.

- **Statement 3 is incorrect:** Placer Deposits are one of the evidences in the support of the Continental Drift Theory. The placer deposits of

gold in the Ghana coast do not have source rock in the region. The gold deposits of Ghana have been derived from the Brazil plateau when the two continents lay side by side.

Q.18) Ans: D

Exp:

- **Statement 1 is correct:** The plate boundaries influence the distribution of natural resources. Many of the Earth's natural resources of energy, minerals, and soil are concentrated near past or present plate boundaries. When the plates move, the oil, gold, and mineral deposits come to the surface from below the surface.
- **Statement 2 is correct:** Gold, as well as other rare metals, can be brought to the surface by plumes of molten rock from deep within the mantle. Water with gold and other metals is heated by magma and forms deposits in volcanoes.

Q.19) Ans: A

Exp:

- **Statement 1 is correct:** Alfred Wegener in his continental drift theory could not explain how continents drifted. The theory of plate tectonics does explain this and states that each plate moves over Earth's hot and semi-plastic mantle. Ocean bottom relief, Paleomagnetic rocks, distribution of earthquakes and volcanoes, gravitational anomalies at trenches, etc. are the evidences put forward by the Plate Tectonics Theory.
- **Statement 2 is incorrect:** Plate tectonics explain the movement of both Continents and Oceans whereas the Continental Drift explains the movement of Continents only. Therefore, the theory of plate tectonics is more complete and it has gained widespread acceptance among scientists.

Q.20) Ans: B

Exp:

- **Statement 1 is correct:** The Mid-Atlantic Ridge runs through Iceland where the ridge is also



known as the Neovolcanic Zone.

- **Statement 2 is correct:** Japan is situated near a destructive plate margin, where the Pacific Plate is being pushed under the Philippine Plate. Japan and earthquakes go hand in hand due to the country's position along the "Pacific Ring of Fire," where it lies across three tectonic plates, including the Pacific Plate under the Pacific Ocean and the Philippine Sea Plate.
- **Statement 3 is incorrect:** The East African Rift Valley is a developing divergent plate boundary in East Africa. Here the eastern portion of Africa, the Somalian plate, is pulling away from the rest of the continent, that comprises the Nubian plate.

Q.21) Ans: B

Exp:

- **Statement 1 is correct:** Metamorphic rocks result from the forces active during plate tectonic processes. The collision of plates, subduction, and the sliding of plates along transform faults create differential stress, friction, shearing, compressive stress, folding, faulting, and increased heat flow. Metamorphic rocks are most abundant at convergent plate boundaries, but can occur in other areas where there are increased pressures and/or temperatures.
- **Statement 2 is incorrect:** Igneous rock forms when magma cools inside the Earth or lava cools above ground and hardens into igneous rocks. The formation of these rocks is the basis of the rock cycle.
- **Statement 3 is correct:** When directed pressure or stress is the dominant agent of metamorphism, it is termed as Dynamic Metamorphism. Dynamic metamorphism is the result of very high shear stress, such as along fault zones. It results mainly from mechanical

deformation with little long-term temperature change.

Q.22) Ans: D

Exp:

- **Statement 1 is correct:** Dharwar Rock System comprises the metamorphic sedimentary rocks of India. The Dharwar rocks are rich in iron ore, manganese, lead, zinc, gold, silver etc.
- **Statement 2 is correct:** The Cuddapah system of rocks have been formed as a result of erosion and sedimentation of the rocks of the Dharwar system. These are also sedimentary rocks.
- **Statement 3 is correct:** The Archean rocks in India are called Purana Rocks/ Pre-Cambrian rocks are the oldest rocks of the earth's crust. Most Archean rocks are igneous in nature. They are found in the Aravalli hills, Deccan plateau and the northeast of India. Because of their volcanic origin, they are crystalline and are foliated.

Q.23) Ans: D

Exp:

- **Statement 1 is incorrect:** The Jaduguda Mine is a uranium mine in Jaduguda village of Jharkhand. It is the largest and first of India's uranium mines in India.
- **Statement 2 is incorrect:** India has only 2% of the world's Uranium reserves. India has uranium reserves in Rajasthan, Jharkhand, Chhattisgarh, Meghalaya, Telangana, Andhra Pradesh and Karnataka. It is currently operating mines in Jharkhand and Andhra Pradesh.

Q.24) Ans: B

Exp:

- **Pair 1 is correct:** Shale is a fine-grained sedimentary rock composed of clay and organic matter. Shale is widespread in nature, making up about 55% of sedimentary rocks. The defining



characteristic of shale is its ability to break into layers or fissility. Shale is commercially important. It is used to make brick, pottery, tile, and Portland cement.

- **Pair 2 is incorrect:** Quartzite is a non-foliated metamorphic rock. Sandstone is converted into quartzite through heating and pressure usually related to tectonic compression within orogenic belts. Quartzite is extremely popular due to its marble-like appearance and granite-like properties.
- **Pair 3 is correct:** Schist is a foliated metamorphic rock. It usually forms on a continental side of a convergent plate boundary where sedimentary rocks, such as shales and mudstones, have been subjected to compressive forces, heat, and chemical activity.

Q.25) Ans: C

Exp:

- **Statement 1 is incorrect:** The arrangement of grains/minerals in the metamorphic rocks is called Foliation. It is caused by the re-alignment of minerals when they are subjected to high pressure and temperature. Foliated metamorphic rocks have a layered or banded appearance. Examples of foliated rocks include: gneiss, phyllite, schist, and slate.
- **Statement 2 is correct:** Sedimentary rocks provide the temporal framework that connects processes within the Earth to those at the surface. These rocks form at temperatures and pressures that do not destroy fossil remnants. And by studying the fossil record one can tell how long life has existed on Earth, and how different plants and animals are related to each other etc. Fossils can tell a lot about the past.
- **Statement 3 is incorrect:** Extrusive Igneous Rocks are characterized by fine-grained textures because their rapid cooling at or near the surface did not provide enough time for large crystals to grow.

Q.26) Ans: D

Exp:

- **Option 1 is correct:** When a rock is subjected to increased temperatures and pressures, certain minerals will undergo chemical reactions and turn into new minerals, while others might just change their shape. This will be dependent upon the Mineral composition of the Parent Rock.
- **Option 2 is correct:** The pressure has the ability to squeeze the spaces out of the minerals within the rock. This makes the rocks denser. The heat and pressure together cause the rock to flow instead of break or fracture.
- **Option 3 is correct:** Water speeds up the process of metamorphism. It also allows for the transportation of ions from one place to another.
- **Option 4 is correct:** The temperature that the rock is subjected to is a key variable in controlling the type of metamorphism that takes place. Rocks of different compositions begin to melt at different temperatures.

Q.27) Ans: B

Exp:

- **Statement 1 is correct:** Basalt is the Mafic rocks that constitutes the igneous oceanic crust. Basalt is created at mid-ocean ridges as a byproduct of partial melt from upwelling mantle.
- **Statement 2 is correct:** During the process of metamorphism the mineral content and texture of the parent rock are changed due to changes in the physical and chemical environment of the rock. Due to heat and pressure exerted on the parent rock, they must adapt to the new conditions and it does so by changing mineral composition and texture.
- **Statement 3 is incorrect:** The most common igneous rock found at the earth's surface is Granite. Granite is a light-colored, coarse grained, intrusive igneous rock as it forms from



the slow crystallization of magma below Earth's surface. It is composed mainly of quartz and feldspar with minor amounts of mica, other minerals etc.

sediment on top of each other/ depositions of sediments over a period of time. The sequence of beds that characterizes sedimentary rocks is called bedding.

Q.28) Ans: A

Exp:

- **Statement 1 is correct:** Bauxite is a sedimentary rock. It is the world's main source of aluminium and gallium. Bauxite is the principal ore of aluminium. The major impurities in Bauxite are iron oxides and hydroxides (which give reddish color to most bauxites) and clay minerals.
- **Statement 2 is incorrect:** Odisha is the largest producer of Bauxite in India. Along with Orissa, Gujarat, Jharkhand, Maharashtra, Chhattisgarh, Tamil Nadu and Madhya Pradesh are the other main bauxite producing states in India.
- **Statement 3 is correct:** Bauxite ore is the world's primary source of aluminum. The Bayer process is the principal industrial means of refining bauxite to produce alumina (aluminium oxide). The Bayer process involves four steps: digestion, clarification, precipitation, and calcination.

Q.29) Ans: B

Exp:

Statement 1 is incorrect: Non-Metallic Minerals do not contain metal content. Sulphur, phosphates and nitrates are examples of non-metallic minerals. However, Aluminium falls under the category of Non ferrous Metallic Minerals.

Statement 2 is correct: The largest reserves of copper ore are in the state of Rajasthan. The major copper mines are the Khetri copper belt in Rajasthan, Singhbhum copper belt in Jharkhand and Malanjkhand copper belt in Madhya Pradesh.

Q.30) Ans: C

Exp:

- **Statement 1 is not correct:** Sedimentary rocks are arranged in layers called beds or strata. Strata is formed by the deposition of layers of

- **Statement 2 is correct:** The major energy resources such as Oil, natural gas, coal, and uranium etc are formed in and come from sedimentary rocks. Sand and gravel for construction come from sediments. Phosphate-bearing sedimentary rocks are used for fertilizer.
- **Statement 3 is correct:** Sedimentary rocks tend to be more porous than igneous rocks because there is more open space between the individual sediment grains than between the minerals in a crystallized igneous rock.

Q.31) Ans: C

Exp:

- **Statement 1 is correct:** Mica is a non-metal and hence a bad conductor of electricity. Due to this property it is widely used as an electric insulator in various electrical and electronic industries. Mica is used in these industries due to its excellent dielectric strength, low power loss factor, insulating properties and resistance to high voltage.
- **Statement 2 is incorrect:** About 95 per cent of India's mica is found in just three states of Andhra Pradesh, Rajasthan and Jharkhand. India is the largest producer and holds the monopoly to the mica production. India produces almost 60% of the world's mica.
- **Statement 3 is correct:** Mica is a naturally occurring mineral dust often used in makeup to add shimmer and sparkle to the cosmetic products. It gives a glowing effect to the skin when it is applied. Mica minerals are often used as color additives in cosmetics.



Q.32) Ans: A

Exp:

- **Statement 1 is correct:** Temperature is a key factor in the formation of both Metamorphic and Igneous rocks. Igneous rocks form when rocks are heated to the melting point which forms magma and Metamorphic rocks form under the action of pressure, volume and temperature (PVT) changes.
- **Statement 2 is incorrect:** Igneous rocks are formed from the cooling and crystallization of the Magma. Whereas ,Metamorphic rocks are formed as a result of temperature and/or pressure action on existing rocks causing changes in the composition and resulting in the appearance of minerals in rocks.

Q.33) Ans: C

Exp:

- **Statement 1 is correct:** Sills are flat intrusion of igneous rock that forms between preexisting layers of rock. They occur in parallel to the bedding of the other rocks.It is concordant with existing layering of rocks.
- **Statement 2 is incorrect:**Stratas are formed during sediment deposition and therefore they are most prominent in the Sedimentary Rocks.However, Stratification also happens in those igneous rocks formed at the Earth's surface,as from lava flows and volcanic fragmental deposits.
- **Statement 3 is correct:** Dikes are tabular bodies of magma that cut through and across the layering of adjacent rocks. They form when magma rises vertically into an existing fracture, or creates a new crack by forcing its way through existing rock, and then solidifies.It is discordant with the existing layers of rocks.

Exp:

- **Option 1 is correct:**
 - **Attrition** is the process of erosion that occurs during rock collision and transportation.
 - It is the wear and tear of transported materials themselves when they collide with each other. The transportation of sediment chips and smoothens the surfaces of bedrock; this can be through water (fluvial) or wind (aeolian). Attrition is also partially responsible for turning boulders into smaller rocks and eventually to sand.
- **Option 2 is incorrect:**
 - In glaciology and meteorology, **ablation**—the opposite of accumulation—refers to all processes that remove snow, ice, or water from a glacier or snowfield. Ablation refers to the melting of snow or ice that runs off the glacier, evaporation, sublimation, calving, or erosive removal of snow by wind.
 - It therefore is a process that carves **Glacial landforms**.
- **Option 3 is incorrect**
 - **Deflation**, in geology, erosion by wind of loose material from flat areas of dry, uncemented sediments such as those occurring in deserts, dry lake beds, floodplains, and glacial outwash plains.
 - It is not related to the Fluvial cycle of erosion.
- **Option 4 is correct**
 - Abrasion is a process of erosion which occurs when material being transported wears away at a surface over time. It is the process of friction caused by scuffing, scratching, wearing down, marring, and rubbing away of materials. Rocks and sediment grinding against each other wear away surfaces. It happens as wind and water rush over rocks. The rocks become smoother as rough and jagged edges break off.

Q.34) Ans: B

Q.35) Ans: D

Exp:

- **Tombolo**

- A tombolo is a sandy **isthmus**. It is a deposition landform by which an island becomes **attached to the mainland by a narrow piece of land** such as a spit or bar. Once attached, the island is then known as a **tied island**. **Tombolos are formed by wave refraction and diffraction.**



- **Spits**

- A spit or sandspit is a **deposition bar or beach landform off coasts or lake shores**. It develops in places where re-entrance occurs, by the process of longshore drift by **longshore currents**
 - The drift occurs due to **waves meeting the beach at an oblique angle, moving sediment down the beach in a zigzag pattern**. This is complemented by longshore currents, which further transport sediment through the water alongside the beach.

Q.36) Ans: B

Exp:



- **Option 1 is incorrect:**

- A **hamada** is a type of desert landscape consisting of **high, largely barren, hard rocky plateaus**, where **most of the sand has been removed by deflation**. The finer-grained products are taken away in suspension, while the sand is removed through saltation and surface creep, leaving behind a landscape of gravel, boulders and bare rock.



- **Option 2 is correct:**

- A playa, is either a **basin or depression that formerly contained a standing surface water body**, which disappeared when **evaporation processes exceeded recharge**. It is a **flat-bottom depression** found in interior desert basins and adjacent to coasts within arid and semiarid regions.



- **Option 3 is incorrect:**

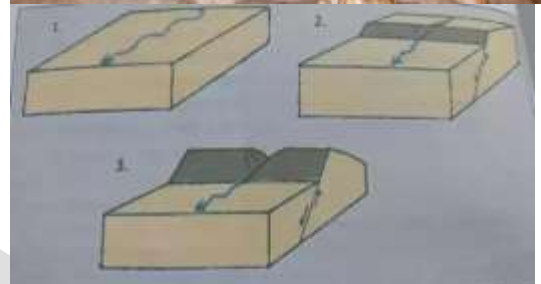
- **Zeugenite** is a **table-shaped area of rock** found in arid and semi-arid areas formed when more resistant rock is reduced at a slower rate than softer

rocks around it under the effects of wind erosion.



• **Option 4 is incorrect:**

- A Yardang is a sharp irregular ridge of sand lying in the direction of the prevailing wind in exposed desert regions, **formed by the wind erosion of adjacent material** which is less resistant.



- **Statement 2 is not correct:** A stalagmite is an upward-growing mound of mineral deposits that have precipitated from water dripping onto the floor of a cave. Most stalagmites have rounded or flattened tips. There are many other types of mineral formations found in caves.

- **Stalagmites** are found in **Mawmluh Cave** in **Meghalaya**, **Kutumsar Caves** at **Kanger Valley National Park** and **Belum caves** in **Andhra Pradesh**

- **Statement 3 is correct:** Coastline of **emergence** is formed either by an **uplift** of the land or by the **lowering** of the sea level. Coastline of **submergence** is an exact opposite case.

- The **east coast of India**, especially its south-eastern part (**TamilNadu coast**), appears to be a **coast of emergence**.
- The **west coast** of India, on the other hand, is **both emergent and submergent**.
- The **northern portion** of the coast is **submerged** as a result of faulting and the southern portion, that is the **Kerala coast**, is an example of an **emergent coast**.

Q.37) Ans: A

Exp:

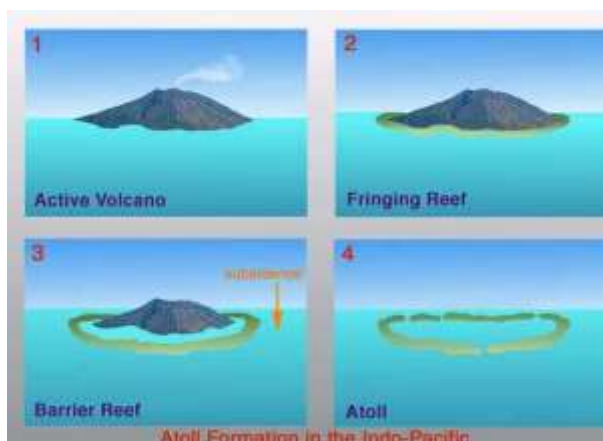
- **Statement 1 is correct:** An **antecedent stream** is a stream that maintains its original course and pattern despite the **changes in underlying rock topography**. A part of a river slope and the surrounding area gets uplifted and the river sticks to its original slope, cutting through the uplifted portion. Examples include **River Ganga, Sutlej and Indus**

- **Statement 4 is incorrect:** Barren Island is an island located in the **Andaman Sea**. It is the **only confirmed active volcano in India and South Asia**.

Q.38) Ans: B

Exp:

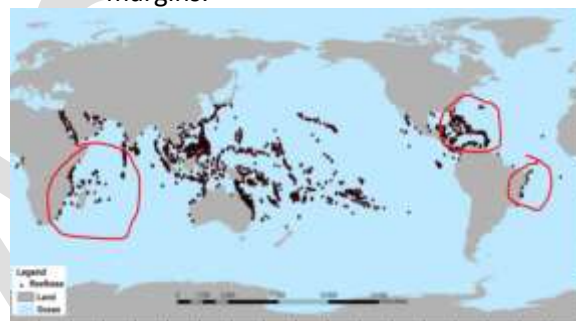
- **Statement 1 is incorrect:**



- **Fringing reefs** grow near the coastline around islands and continents. They are separated from the **shore by narrow, shallow lagoons**. Fringing reefs are the most common type of reef that we see.
 - **Barrier reefs** also parallel the coastline but are separated by deeper, wider lagoons. At their **shallowest point, they can reach the water's surface** forming a "barrier" to navigation. **The Great Barrier Reef in Australia is the largest and most famous barrier reef in the world.**
- **Statement 2 is correct:** **Atolls** are rings of coral that create protected lagoons and are usually located in the middle of the sea. Atolls usually form when islands surrounded by fringing reefs sink into the sea or the sea level rises around them (these islands are often the tops of underwater volcanoes). The fringing reefs continue to grow and eventually form circles with lagoons inside.
- **Statement 3 is correct:** Reef-building corals cannot tolerate water temperatures below 18° Celsius. Many grow optimally in water

temperatures between 23°–29°Celsius. This Virtually limits the areal distribution of corals to the tropical and subtropical zones. They are averse to **cold currents** because of the **upwelling of cold water from the depths that cools the warm water surface**. The **Western margins of the continents generally receive cold water currents** from **higher latitudes**. Coral reefs are thus **usually absent in the Western margins** of continent

- As can be seen in the map, most coral reefs are found on the eastern margins.

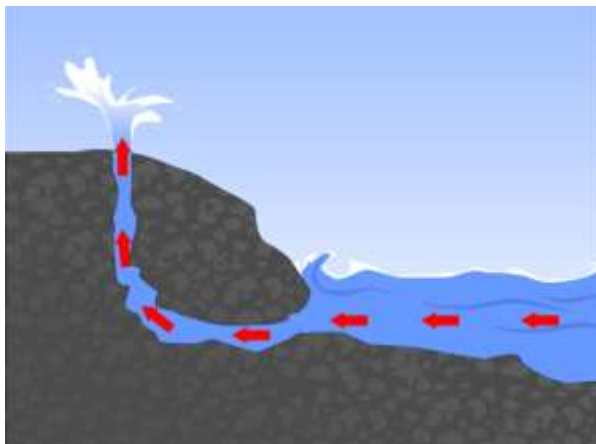


Q.39) Ans: B

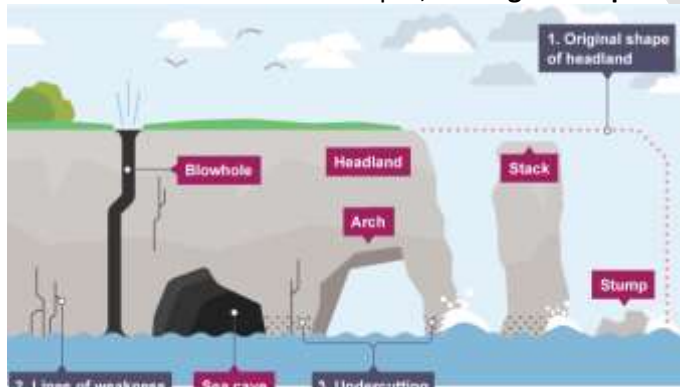
Exp:

Groups, stacks and stumps are features **found near the coast or in the middle of the ocean**. Therefore option **B is correct**.

- **Groups (or Blowhole)** : it is an **opening in the roof of a sea cave** through which **incoming waves may force air to rush upward or water to spout intermittently**. The splashing of waves against the roof of a cave **enlarges the joints when compressed air is trapped inside**. A natural shaft is formed which may pierce through to the surface may spray water out of this.



- **Stacks and stumps:** A stack or sea stack is a geological landform consisting of a steep and often vertical column or columns of rock in the sea near a coast, formed by wave erosion. They are formed when part of headland is eroded by hydraulic action
- Erosion causes the arch to collapse, leaving the pillar of hard rock **standing away from the coast—the stack**. Eventually, erosion will cause the stack to collapse, **leaving a stump**.

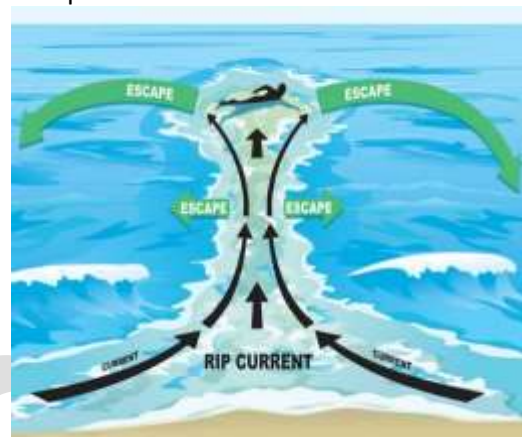


Q.40) Ans: D

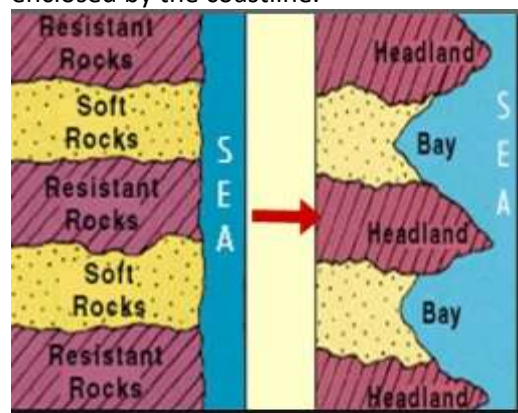
Exp:

- **Statement 1 is incorrect:** Tides are very long-period waves that move through the oceans in response to the forces exerted by the **earth, moon and sun**. Tides originate in the oceans and **progress toward the coastlines where they appear as the regular rise and fall of the sea surface**.
- **Statement 2 is incorrect:** When waves travel from deep to shallow water, they break near the shoreline and generate currents that are influenced by the shape of the ocean floor, at times producing rip currents. Waves

interacting with each other can also cause short-lived rip currents.



- It is a narrow current of water which moves **directly away from the shore**, cutting through the **lines of breaking waves** like a river running out to sea
- Rip currents can be hazardous to people in the water. Swimmers who are caught in a rip current and who do not understand what is happening, or who may not have the necessary water skills, may panic, or they may exhaust themselves by trying to swim directly against the flow of water.
- **Statement 3 is correct:**
 - A gulf is a large **inlet from the ocean into the landmass**, typically with a **narrower opening than a bay**. It traditionally refers to highly-indented navigable bodies of saltwater that are enclosed by the coastline.



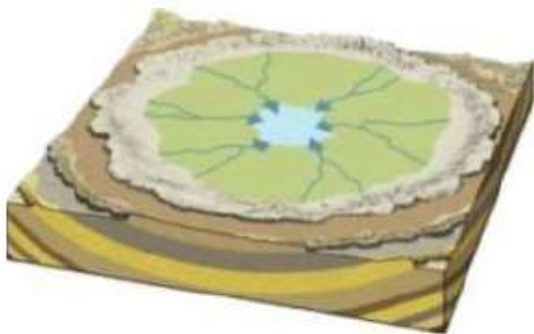
- The geology of the area where a bay is formed, typically consists of alternate bands of hard and soft rocks which allows the sea water to erode

the soft rocks and make inroads towards the mainland.

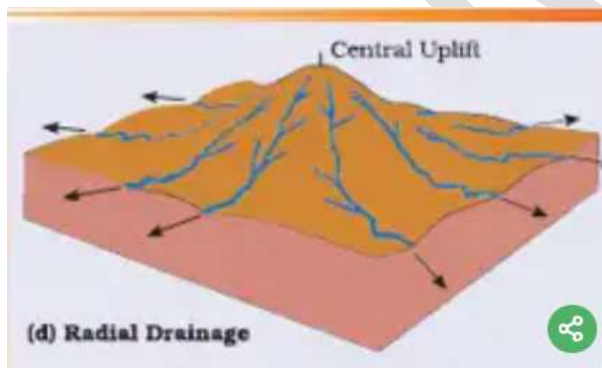
Q.41) Ans: D

Exp:

- **Statement 1 is incorrect:** In a **Centripetal Drainage system**, the streams converge from all sides on to a low lying basin.



- Examples: streams of Ladakh, Tibet, and the Bagmati and its tributaries in Nepal.
- **Statement 2 is Incorrect:** In a **Radial drainage system**, The tributaries from a summit follow the slope downwards and drain down in all directions.



- Examples: Streams of Saurashtra region, the rivers originating from the **Amarkantak Mountain**, Central French Plateau, Mt. Kilimanjaro.
 - The **Narmada** and **Son** originate from **Amarkantak Hills** and flow in different directions.



- **Statement 3 is Incorrect:** The Narmada river is one of the rivers in India that flows in a **rift valley**, flowing **west between the Satpura and Vindhya ranges**, having risen from the **Amarkantak Hills**. Being the rift valley river, Narmada river does not form delta.
- Rift valley river forms estuaries. The other rivers which flow through the rift valley include **Damodar River in Chota Nagpur Plateau** and **Tapti**

Q.42) Ans: D

Exp:

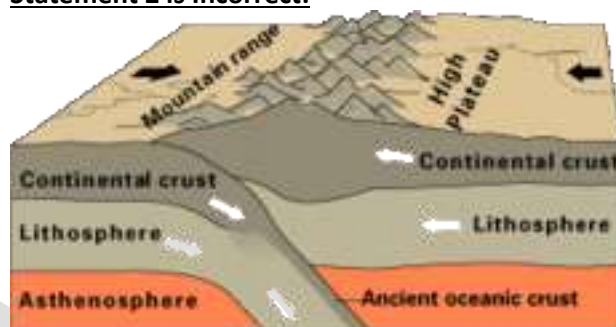
- **All options are correct:**
 - **Diastrophism** is the process of **deformation** of the Earth's crust which involves folding and faulting. Diastrophism can be **considered** part of **geotectonics**. The study of diastrophism **encompasses the varying responses of the crust to tectonic stresses**. These responses include **linear or torsional horizontal movements** (such as continental drift) and vertical subsidence and uplift of the **lithosphere (strain)** in response to natural stresses on Earth's **surface** such as the **weight of mountains, lakes, and glaciers**
 - **Orogeny**, mountain-building event, generally one that **occurs in geosynclinal areas**. In contrast to **Epeirogeny**, an **orogeny tends to occur during a relatively short time in linear belts and results in intensive deformation**.

Orogeny is usually accompanied by folding and faulting of strata, development of angular unconformities (interruptions in the normal deposition of sedimentary rock).

- An **earthquake** is the shaking of the surface of the Earth resulting from a sudden release of energy in the Earth's lithosphere that creates seismic waves.
- The theory of **plate tectonics** states that the Earth's solid outer crust, the lithosphere, is separated into plates that move over the **asthenosphere**, the molten upper portion of the **mantle**. Oceanic and **continental** plates come **together, spread apart, and interact** at **boundaries** all over the planet.

formed when the **faults have a large vertical displacement**. Adjacent **raised blocks (horsts)** and down-dropped blocks (**grabens**) can form high escarpments. Often the movement of these blocks is accompanied by tilting, due to compaction or stretching of the crust at that point.

- **Statement 2 is Incorrect:**

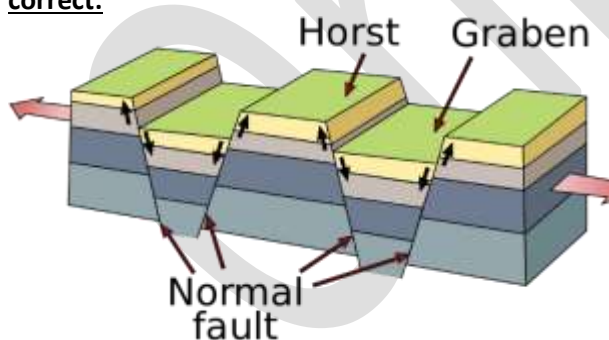


- **Continental-continental convergence**
 - Fold mountains are created where two or more of Earth's tectonic plates are pushed together. At these colliding, **compressing boundaries**, rocks and debris are warped and folded into rocky outcrops, hills, mountains, and entire mountain ranges.
 - Hence **Compression forces** (and not tension forces) are responsible for folding.

Q.43) Ans: C

Exp:

- Statements 1 and 3 are correct:



- **Fault blocks** are very large blocks of **rock**, sometimes hundreds of kilometres in extent, created by tectonic and localized stresses in Earth's crust. **Large areas of bedrock are broken up into blocks by faults**. Blocks are characterized by relatively uniform lithology. The largest of these fault blocks are called crustal blocks
- **Landforms** (mountains, hills, ridges, lakes, valleys, etc.) are sometimes

Q.44) Ans: C

Exp:

- **Mesas**
 - A **mesa** is an **isolated, flat-topped elevation, ridge or hill**, which is bounded from all **sides by steep escarpments** and stands **distinctly above a surrounding plain**. Mesas characteristically **consist of flat-lying soft sedimentary rocks** capped by a **more resistant layer or layers of harder rock**, e.g. **shales** **overlain by sandstones**.



- The **resistant layer acts as a caprock that forms the flat summit of a mesa.** The caprock can consist of either **sedimentary rocks such as sandstone and limestone; dissected lava flows; or a deeply eroded duricrust.** Unlike plateau, whose usage does not imply horizontal layers of bedrock, the **term mesa applies exclusively to the landforms built of flat-lying strata.** Instead, flat-topped plateaus are specifically known as tablelands.

Q.45) Ans: D

Exp:

- **All options are correct:**
 - The lava that is released during volcanic eruptions on cooling develops into igneous rocks. The cooling may take place either on reaching the surface or also while the lava is still in the crustal portion. The lava that cools within the crustal portions assumes different forms. These forms are called intrusive forms. They are:
 - **Batholiths**
 - A large body of magmatic material that cools in the deeper depth of the crust develops in the form of large domes. They appear on the surface only after the denudational processes remove the overlying materials.
 - **Lopolith**

- As and when the lava moves upwards, a portion of the same may tend to move in a horizontal direction wherever it finds a weak plane. It may get rested in different forms. In case it develops into a **saucer shape, concave to the sky body, it is called lopolith.**

○ Sills

- The near horizontal bodies of the intrusive igneous rocks are called sill or sheet, depending on the thickness of the material. The thinner ones are called sheets while the **thick horizontal deposits are called sills.**

○ Dykes

- When the lava makes its way through **cracks and the fissures developed in the land, it solidifies almost perpendicular to the ground.** It gets **cooled in the same position to develop a wall-like structure.** Such structures are called **dykes.**

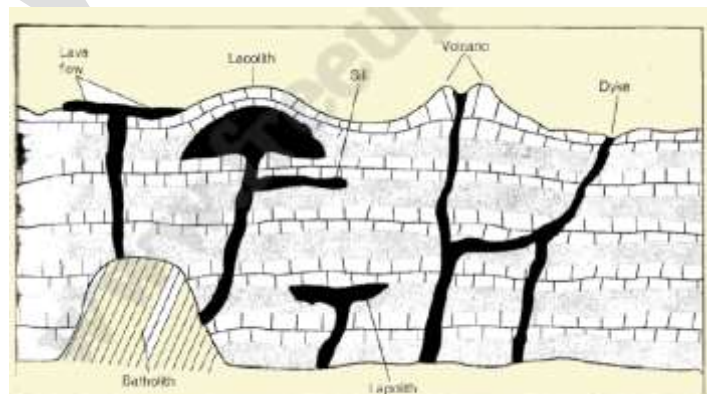


Fig. 7.22: Features associated with intrusive volcanicity

Q.46) Ans: D

Exp:

- Valleys start as small and narrow rills (a rill is a shallow channel cut into soil by the erosive action of flowing water). The rills will gradually develop into long and wide gullies; the gullies will further deepen, widen and lengthen to give rise to valleys. Depending upon

dimensions and shape, many types of valleys like V-shaped valley, gorge, canyon, etc. can be recognised.

- **All statements are correct:**



- A gorge is a **deep valley with very steep to straight sides**. It is almost equal in width at its top as well as its bottom.
- A canyon is **characterised by steep step-like side slopes** and can be as **deep** as a **gorge**. A canyon is wider at its top than at its bottom.
- In fact, a **canyon is a variant of gorge**. Valley types depend upon the type and structure of rocks in which they form. For example, **canyons commonly form in horizontal bedded sedimentary rocks** and gorges form in hard rocks.

Q.47) Ans: B

Exp:

- **Statement 1 is correct:** The weathering of rocks is a natural phenomenon which leads to the enrichment of various metal ores such as iron, manganese, aluminium, copper etc by leaching unwanted minerals leaving behind the valuable ones.
- **Statement 2 is incorrect:** The Chemical weathering of rocks does not cause physical damage. It is a reaction between the chemical composition of the rock and outside chemicals. Chemical weathering can make a rock more vulnerable to physical weathering forces. Chemical weathering also helps the physical weathering by weakening the components of the rocks. This makes the rocks easier to break up by mechanical means.

Q.48) Ans: C

Exp:

- **Statement 1 is incorrect:** Mudflows occur mostly in mountainous places where a long dry season is followed by heavy rains. The Fast-moving flows of mud and rock, called mudflows. Their high speeds and the destructive volume of the mudflows make them dangerous to life and property.
- **Statement 2 is correct:** A rock slide is a type of landslide which occurs when a mass of rock moves quickly downslope. They occur in the mountainous regions or where excavation activities are undertaken such as mining and quarrying etc. Deep excavation of earth loosens the land which is the cause of landslides in excavation sites. A rockslide can also occur after a heavy rainfall.
- **Statement 3 is correct:** A mass movement can occur any time a slope becomes unstable. Undercutting of steep slopes causes instability of the slopes.

Q.49) Ans: C

Exp:

- **Statement 1 is correct:** Solifluction is the flow of water-saturated soil down a steep slope. Solifluction is common in Earth's cold regions. It happens during the summer thaw when the water in the soil is trapped there by frozen permafrost beneath it. Because permafrost is impermeable to water, soil overlying it may become oversaturated and slide downslope under the pull of gravity.
- **Statement 2 is correct:** Landslides can cause seismic disturbances and landslides can also result from seismic disturbances. Example- Earthquakes cause obvious disruption to local substrates that can weaken the resisting forces and trigger landslides.

Q.50) Ans: D



Exp:

- **Pair 1 is correct:** Explosive eruptions are favored by high gas content and high viscosity. Rhyolitic Magma is extremely viscous, due to its high silica content. This favors explosive eruptions. Rhyolitic magma mostly erupts as pyroclastic rock rather than as lava flows.
- **Pair 2 is incorrect:** Shield volcanoes are formed by basaltic magma-low viscosity basaltic magma that flows easily down slope away from the summit vent. Shield volcanoes are composed almost of relatively thin lava flows built up over a central vent.
- **Pair 3 is correct:** Basaltic magma has low viscosity due to its relatively low silica content enabling it to rapidly move as lava flows that can spread over long distances. Example- Hawaii's volcanoes.

Q.51) Ans: C

Exp:

- **Statement 1 is correct:** Karst topography refers to the natural features produced due to the chemical weathering or slow dissolving of limestone, dolostone, gypsum etc. It results from the excavating effects of underground water on massive soluble limestone. It is characterized by underground drainage systems with sinkholes and caves.
- **Statement 2 is incorrect:** There are differences in geothermal gradients and heat flow from within the earth and the strength and crustal thickness, therefore, the action of endogenic forces are uneven.
- **Statement 3 is correct:** Soil is formed through the process of rock weathering. Chemical weathering is the first stage in the production of soils. Examples of chemical reactions leading to weathering are hydrolysis, carbonation, oxidation and hydration. Along with the chemical

weathering, the physical weathering of rocks also assists in the soil formation.

Q.52) Ans: A

Exp:

- **Statement 1 is correct:** Chemical weathering proceeds more slowly in deserts compared to more humid climates because of the lack of water.
- **Statement 2 is incorrect:** Diastrophism is an endogenic geomorphic process. And Orogeny is an important component of Diastrophism. Orogeny is an event that leads to both structural deformation and compositional differentiation of the Earth's lithosphere (crust and uppermost mantle) at convergent plate margins.

Q.53) Ans: D

Exp:

- **Statement 1 is correct:** Erosion affects waterways, rivers, streams etc. Due to erosion the eroded particles deposit in river beds, banks. The excess sedimentation can clog the rivers and streams and causes decline in fish and other species.
- **Statement 2 is correct:** Soil erosion inhibits the spoils capacity to grow nutritious food. By decreasing the nutrients available to plants, loss of rooting depth, degradation of soil structure, decrease in plant-available water reserves, reduction in organic matter, and nutrient imbalance etc, soil erosion can decrease crop yields by up to 50 percent. In addition, crops that do grow tend to be of a lower quality: misshapen, smaller and less nutritious.
- **Statement 3 is correct:** Soil erosion helps change the landscape of any geographic area. Landforms can be created and destroyed by erosion. Many landscapes across the world are formed from the rapid rushing of water. Example-The Grand Canyon. The action of



erosion can create an array of coastal landscape features. Example- Sea Stacks.

Q.56) Ans: C

Exp:

Q.54) Ans: B

Exp:

- **Option 1 is incorrect:** Sheet erosion occurs when a thin layer of topsoil is removed by rainwater that does not infiltrate into the ground. It occurs in a wide range of settings such as coastal plains, hillslopes, floodplains and beaches.
- **Option 2 is correct:** Gully erosion is the process by which gullies/deep channels are formed. The running water cuts through the clayey soil and makes deep channels in it. As a result of this, the land becomes unfit for cultivation. Hillsides are more prone to gully erosion when they are cleared of vegetation.
- **Option 3 is incorrect:** Rill erosion is a form of water erosion in which the erosion takes place through numerous narrow and zig-zag channels. It is caused by slow movement of water along small channels on bare land with less vegetative cover.

- **Statement 1 is incorrect:** The orogenic processes involve mountain building through severe folding and faulting of the earth's crust. Whereas, the Epeirogenic processes involve uplift or warping of large parts of the earth's crust. Orogeny leads to both structural deformation and compositional differentiation of the Earth's lithosphere. Epeirogenic processes lead to the formation of continents.
- **Statement 2 is correct:** Diastrophic movements are gradual and might stretch for thousands of years. They cause changes very gradually which might not be visible during a human lifetime.
- **Statement 3 is incorrect:** Endogenic forces or endogenetic forces are the pressure that originates inside the earth, whereas, the Exogenic forces are usually driven by gravity or atmospheric forces.

Q.55) Ans: B

Exp:

- **Statement 1 is correct:** Salt weathering is a form of mechanical/physical weathering of rock. It is mainly due to the growth and expansion of various salts crystals resulting in the physical disintegration of rocks. Crystallisation of salts causes a dramatic increase in volume which exerts pressure on the surrounding rock, and can eventually fracture the rock.
- **Statement 2 is incorrect:** Hydration is a form of chemical weathering in which the chemical bonds of the mineral are changed as it interacts with water. This process is reversible and long in nature. The continued repetition of this process causes fatigue in the rocks and may lead to their disintegration.

Q.57) Ans: D

Exp:

- **Pair 1 is correct:** The sudden expansion and contraction of rocks results in the peeling off of the outer layers of the rocks. This process of weathering is known as Exfoliation. It can occur due to expansion and contraction induced by temperature changes, Unloading or release of stress in a rock that produces expansion joints etc.
- **Pair 2 is correct:** Abrasions are caused by the frictional force impacting upon the rocks. It is a type of Mechanical Weathering. The rocks and sediment grinding against each other wear away surfaces. The intensity of abrasion depends on the hardness, concentration, velocity and mass of the moving particles.
- **Pair 3 is correct:** Stalactites and stalagmites are formed by chemical weathering. Both are the natural formations found in the Karst



topography. Stalactites grow down from the cave ceiling, while stalagmites grow up from the cave floor.

Q.58) Ans: B

Exp:

- **Statement 1 is incorrect:** Topography is a passive control factor in the Soil formation Process. Soils formed on the hill sides are shallow, due to erosional losses. Soils in the valleys tend to be deeper, darker, and contain more horizons. Gentle slopes are favourable for the soil formation.
- **Statement 2 is correct:** Dead plants provide the organic matter to the soil. Some organic acids which form during humification aid in decomposing the minerals of the soil parent materials. Plants also cause chemical weathering. As plant roots take in nutrients, they remove elements from the minerals. This causes a chemical change in the rock which helps in the weathering of rocks and ultimately in soil formation.
- **Statement 3 is correct:** The redistribution of topsoil and associated nutrients is caused due to soil erosion. This redistribution is very essential for the biogeochemical cycles and other nutrient cycles.

Q.59) Ans: A

Exp:

- **Pair 1 is correct:** Yardangs are formed by wind erosion through the process of abrasion. They are formed on vertical hard/soft layers of rock. The rock is eroded into alternating ridges and furrows essentially parallel to the dominant wind direction. The direction of the yardangs can indicate the direction of the prevailing wind.
- **Pair 2 is incorrect:** U-shaped valleys are glacially-carved valleys. U-shaped valleys are related to Glaciers, whereas the V-shaped valleys are carved by rivers.

- **Pair 3 is incorrect:** Rivers are associated with Alluvial Deltas. They are depositional landforms of the rivers. Deltas are formed near the mouth of streams where it meets seas, lakes or oceans.

Q.60) Ans: D

Exp:

Statement 1 is correct - A blood moon occurs during a full eclipse, when the moon is fully in Earth's shadow. A little bit of light from Earth's sunrises and sunsets falls on the surface of the moon. When this red light strikes the moon's surface, it also appears red and hence is called a blood moon.

Statement 2 is incorrect - A supermoon is a larger than usual moon, because it is closer to the Earth. It is also called the Perigean full moon. Apogee is when the moon is farthest from the Earth while Perigee is the opposite.

Statement 3 is correct - It is not a moon with Blue glow or colour. The moon on this event looks just like it would on any other full moon. *The only thing special about it is that it is one extra full moon in a season.*

Statement 4 is correct - The full moon that happens nearest to the autumnal equinox (usually September 22nd or 23rd) takes the name "harvest moon".

Extra edge:

Important facts about Blood moon and half-moon

- The last blood moon on Jan. 20-21, 2019 coincided with a supermoon and the Full Wolf Moon, earning it the title "Super Blood Wolf Moon."
- "Half-blood moon" is a partial lunar eclipse.

Source: <https://www.space.com/39471-what-is-a-blood-moon.html>

UPSC Q. referred: Q.51 pre 19 set A

Q.61) Ans: A

Exp:



Statement 1 is correct: An image of the supermassive black hole and its shadow at the center of a galaxy known as M87 has been captured in 2019 for the first time.

Statement 2 is correct: Sagittarius A is a black hole with about 4 million times the mass of the Sun. It is just 26,000 light years from Earth.

Statement 3 is incorrect: The Sun would need to be about 20 times more massive to end its life as a black hole.

Extra edge:

Stars and Black holes

- Not all stars become black holes. Some smaller stars are big enough to go supernova, but too small to become black holes — they'll collapse into super-dense structures called **neutron stars**.
- But the Sun's not big enough for this fate, either. It is also called a yellow dwarf.

Source: [Why the Sun Won't Become a Black Hole](#)

UPSC Q. referred: Pre 19 Q.51 set A

Q.62) Ans: B

Exp:

Statement 1 is correct - Electric currents inside the sun generate a magnetic field that spreads throughout the solar system. The field causes activity at the surface of the sun, surging and ebbing in a regular cycle. At the peak of the cycle, the polarity of the field flips, during a time of maximum sunspot activity.

Statement 2 is incorrect - Sunspots are areas that appear dark on the surface of the Sun, because they are **cooler** than other parts of the Sun's surface.

Statement 3 is correct - They carry Sun's magnetic waves along and hence can damage the satellites and electronic equipment, Global Navigation Satellite System (GNSS) and create harmful geomagnetic induced currents (GICs) in the power grid and pipelines.

Extra edge by Only IAS

Parker Solar probe mission

- The spacecraft is flying closer to the Sun's surface than any previous spacecraft. In 2025, the spacecraft will complete 24 orbits around the Sun. The spacecraft will come within about 3.9 million miles (6.3 million km) of the Sun's surface.

Source: [NASA/Marshall Solar Physics](#), Indian Express explained section.

UPSC Q. referred: Pre 2020 set B Q.59

Q.63) Ans: B

Exp:

Option 2 is correct - "ACCRETION BURST EVENT" happens when a forming star swallows a huge amount of matter resulting in a burst of activities in the star. It can help us understand fully the process of formation of stars and how they gain mass.

Source: [Astronomers Have Caught a Rare And Massive 'Accretion Burst' in Our Galaxy](#)

UPSC Q. referred: Pre 2019 set A Q. 44

Q.64) Ans: C

Exp:

Statement 1 is incorrect - An aurora is a colourful display of lights predominantly seen in HIGHER latitudes. They are observed as "Aurora Borealis" near the magnetic North pole region and "Aurora Australis" near the magnetic south pole region.

Statement 2 is correct - It is possible in case of a very strong geomagnetic storm or major solar flares. In 2001 they were observed from as low as Southern California due to a major geomagnetic storm.

Statement 3 is correct - Auroras occur due to interaction of solar wind and magnetosphere of Earth.



Due to solar flare emerging from a sunspot, or CME, this becomes possible almost anywhere the solar wind touches the magnetosphere. Even outside the magnetic North and South pole.

Source: <https://www.forbes.com/sites/jamiecartereurope/2020/09/28/northern-lights-possible-over-us-this-week-as-strong-geomagnetic-storms-predicted-say-scientists/?sh=7b3c0a6e65f7>

UPSC Q. referred: Pre 2020 set B Q. 59

Q.65) Ans: B

Exp:

Statement 1 is correct: The rivers carry materials from land to the sea which settle overtime. Thus a layer of sediments and silt is deposited on a floor which came into existence due to seafloor spreading.

Statement 2 is correct: Within the top layer of mud and silt, there are rare earth and other valuable minerals like nickel and cobalt. These, if mined, can provide us with raw materials to sustain our energy needs for centuries to come.

Statement 3 is incorrect: The abyssal plains are covered with smooth terrigenous deposits and not coarse materials.

Source: Physical geography NCERT

UPSC Q. referred: Pre 2019 Q.68 set A

Q.66) Ans: C

Exp:

Statement 1 is Correct: The ocean floor can be divided into 4 main divisions- continental shelf, continental slope, deep sea plains and ocean deep. Apart from these divisions there are also major and minor relief features in the ocean floors like ridges, hills, sea mounts, guyots, trenches, canyons, etc.

Continental shelf is an extended margin of each continent occupied by relatively shallow seas and gulf. It is the shallowest part of the ocean showing an

average gradient of 1° or even less. The shelf typically ends at a very steep slope, called the shelf break. The width of the continental shelves vary from one ocean to another.

Statement 2 is Correct: Most continental shelves are broad, gently sloping plains covered by relatively shallow water. Thus, Sunlight penetrates the shallow waters, and many kinds of organism's flourish. Further, Ocean currents and runoff from rivers bring nutrients to organisms that live on continental shelves. Plants and algae make continental shelves rich feeding grounds for sea creatures.

Q.67) Ans: c

Exp:

Statement 1 is Correct: It stretches nearly 65,000km long with more than 90 percent of it lying in the deep ocean. It is most well developed in the Atlantic Ocean running in an s-shape in N-S direction.

Statement 2 is incorrect: The MOR's/submarine ridges are located at the juncture of the divergent boundary and not the convergent ones. These are places where new ocean floors are created due to seafloor spreading. (Divergent=constructive boundary. While convergent=destructive boundary)

Statement 3 is correct: Plate Tectonics theory tries to explain most of the dynamism of the earth's crust and features of the endogenetic forces. The mid oceanic ridges are evidence in support of the theory as through their study we can find out the age of the oceanic crust, earth's magnetic orientation in the past etc.

Statement 4 is incorrect: The longest MOR is in the Atlantic Ocean and not in the Pacific. The one in the Atlantic is so large that it can be seen from space.

Source: Geography NCERT

Q.68) Ans: B

Exp:

Pair 1 is correct - MOR's can have peaks as high as 2500 m and some reach even above the ocean surface. Iceland is a part of the Mid oceanic ridge. They are the longest running mountain chains on our planet (around



95000 km). The largest one is found in the Atlantic ocean and runs in N-S direction in an S-shape.

Pair 2 is Incorrect - Seamounts are mountains with pointed summits rising from the seafloor that does not reach the surface of the ocean. Seamounts are volcanic in origin. An example is the emperor seamount chain in the Pacific Ocean.

Pair 3 is Incorrect - Guyots are flat topped seamounts with evidence of gradual subsidence. They have evidence which suggests that they were once above the sea surface and are now submerged.

Pair 4 is correct - Submarine canyon: These are deep valleys sometimes found cutting across the continental shelves and slopes, often extending from the mouths of large rivers. Hudson Canyon is the best known example.

Source: Geography Ncert

Q.69) Ans: C

Exp:

Statement 1 is Correct: They are also called “ocean deeps” and are the deepest parts of the oceans. They may/may not always be in the middle of the ocean. For example, the Philippines trench, Kuril-Kamchatka trench etc are closer to land.

Statement 2 is Correct: Peru-Chile Trench/Atacama trench running parallel to the Andes, Java trench etc are examples of such trenches. These occur at places where convergent plate boundaries are interacting and one is going under the other.

Statement 3 is Incorrect: The largest number of ocean trenches are in the Pacific Ocean. This has something to do with the “Pacific ring of fire” and the volcanism surrounding it.

Q.70) Ans: B

Exp:

Statement 1 is incorrect: Pelagic deposits can consist of both organic and inorganic material. They are fine grained sediments found on the ocean floor usually far away from land. The organic material is usually the

ooze. It is kind of a liquid mud, made up largely of the remains of various marine organisms.

Statement 2 is correct: The red clay that usually originates from volcanic activity makes up most of the inorganic material. Red clay is mainly made up of silicon and aluminium dioxide, while the other constituents can include radium, phosphorus, manganese and iron.

Extra Edge by Only IAS

Ocean

deposits

- According to the characteristics of ocean depth, ocean deposits can be classified as pelagic deposits, and terrigenous deposits.
- While terrigenous deposits are found mostly at the shallower depths can include deposits of mud, sand, gravel, and volcanic materials, derived from denudation of continental rocks,
- Pelagic deposits can consist of organic material in the form of marine plants and animals as well as inorganic material.
- Pelagic deposits can cover about 75.5 percent of ocean areas in the form of many types of oozes in a sporadic mix in most cases with other types of ocean deposits.

Source: Geography class 11th Ncert

UPSC Q. referred: Q.53 set B Pre 2020

Q.71) Ans: B

Exp:

- **Statement 1 is incorrect:** Cloudbursts are the events of sudden flash rains over a short period of time. As per **IMD**, cloudbursts occur when more than 100 millimeters of rainfall occurs over a region of 20-30 sq. km within an hour. It occurs when moisture-carrying winds move up the hilly terrain forming cumulonimbus clouds accompanied with orographic lift. It occurs mostly in the **hilly terrain** however; instances of cloudbursts are also found in the plain areas.
- **Statement 2 is correct:** Cloudbursts occur with the formation of **cumulonimbus clouds**.

Cumulonimbus clouds are formed at a height of 4-7 km and appear like cotton wool.

- **Statement 3 is correct:** **Copenhagen climate adaptation plan** aims at providing strategies for dealing with extreme events of cloudbursts. It aims to provide strategies for climate change.
- **Statement 4 is incorrect:** They are created through the **upward movement** of air currents forming cumulonimbus clouds usually in the areas of tropical cyclones. The rain drops are carried upwards and grow heavier in size. After crossing a certain threshold they come down as heavy rains.

Q.72) Ans: B

Exp:

- **Statement 1 is incorrect:** **Zoonotic diseases** are those which are transferred from animals to humans through pathogens like **bacteria, viruses, parasites or fungi**. These diseases can lead to significant disruptions in the supply of the animal food and other animal products and account for 75% of all the new and emerging diseases as per the **National Centre for Disease Control (NCDC)**. Climate change, absence of hygiene, mutations, etc. are some of their causes for emergence. **Japanese Encephalitis, Nipah Virus, Ebola Virus, COVID-19**, etc. are some of the examples of the zoonotic diseases.
- **Statement 2 is incorrect:** **E-2025** initiative was launched by the **World Health Organization** during the **World Malaria Day**. It is a group of 25 countries that are technologically competent to eliminate malaria within the next 5 years. Malaria is caused by **Plasmodium** parasites and is transmitted to humans through the bite of the **female Anopheles mosquito**. India has launched several programmes to deal with malaria:
 - **National Vector Borne Disease Control Programme**, that includes control of six vector borne diseases including that of Malaria.
 - **National Framework for Malaria Elimination 2016- 2030**.
- **Statement 3 is correct:** **Sickle cell anemia** is an inherited red blood cell disorder which is also a rare or an orphan disease. In this disease the

shape of the normal red blood cells is transformed from the normal disc shaped to that of the crescent or sickle shaped and affects the supply of oxygen to the rest of the body, in effect affecting the hemoglobin. It is a rare disease with no possible cure except for the bone marrow transplant.

Q.73) Ans: C

Exp:

- **Statement 1 is incorrect:** **Pradhan Mantri Garib Kalyan Ann Yojana** aims at providing 5 kg free rice/wheat per person per month to the beneficiaries of the National Food Security Act. It is given in addition to the existing monthly Quota under the National Food Security Act. It was launched as a component under **Atma Nirbhar Bharat** to ensure food security to the poor and migrants affected by the COVID induced lockdown.
- **Statement 2 is incorrect:** The entire cost for the supply of the foodgrains is borne totally by the Central government.
- **Statement 3 is correct:** The scheme under the **Atma Nirbhar Bharat** is touted as the world's biggest food security programme covering a beneficiary of around 80 crores.

Q.74) Ans: A

Exp:

- **Statement 1 is incorrect:** **India H₂ Alliance** is an initiative by the **Chert Industries and the Reliance Industries Ltd**. The aim of the project is to create an enabling environment with the enhancement of opportunities in the hydrogen economy and supply chain. The alliance would be a collaboration of the **private sector, government and the public**, and would ensure using the hydrogen for the net reduction in the carbon emissions.
- **Statement 2 is correct:** The **India H₂ alliance** would offer help and expertise to the government in the development of the **National Hydrogen Policy and Roadmap by 2030** along with creation of **National Hydrogen Fund and the National Hydrogen Taskforce and Mission**.

Q.75) Ans: D



Exp:

- **Statement 1 is incorrect:** 8th schedule of the Constitution deals with the official language or the schedule languages. Originally, there were 14 in number while later increased to 22. The constitution does not explicitly mention the criteria for the inclusion of the language in the schedule. It is the **Commission on Linguistic minorities** that decides the language to be added followed by the Constitutional amendment by the Central government.
- **Statement 2 is incorrect:** Article 29 deals with the conservation of language, script, or culture by any section of citizens residing in any part of India. It is applicable to all citizens irrespective of whether they are in minority or not. Article 30 on the other hand deals with the rights of the minorities to establish and administer educational institutions of their choice. It applies to both linguistic and religious minorities.
- **Statement 3 is incorrect:** Any change in the 8th Schedule requires an amendment to the Constitution by the Central Government and does not involve the governments of the states. The **21st Constitutional Amendment Act, 1967** added Sindhi to the 8th schedule. The **71st Constitutional Amendment Act, 1992** added three languages (Konkan, Nepali, and Manipuri) while the **92nd Constitutional Amendment Act, 2003** added four languages (Bodo, Dogri, Maithili, and Santhali).

Q.76) Ans: A

Exp:

- **Statement 1 is correct:** Nancowry group of Islands are associated with the **Andaman and Nicobar Islands in India.** Car Nicobar and Nancowry have recently become the first largest and contiguous territory to be conferred with organic certification.
- **Statement 2 is incorrect:** La Soufriere is a Volcano located in the Caribbean Islands in the state of **St. Vincent and the Grenadines.** It is a stratovolcano which recently erupted a large amount of Sulphur dioxide in the atmosphere.
- **Statement 3 is incorrect:** Plain of Jars, located in **Laos**, in the plateau of Xiangkhoang. It became a UNESCO world heritage site in 2019

and is a megalithic archaeological landscape consisting of a number of stone jars. Laos is the only landlocked country in Southeast Asia with its capital at Vientiane.

Q.77) Ans: C

Exp:

- **Statement 1 is incorrect:** **World Trade Organization (WTO)** is a global organization that deals with the conduct of trade among the member nations with its headQ.uarters in **Geneva.** Its **Agreement on Agriculture** deals with reduction in trade barriers by reducing domestic support, providing market access, and export subsidy. The agriculture subsidy under this is calculated as a percentage of the total value of production of a crop.
- **Statement 2 is correct:** **Green box subsidies** involve the subsidies that do not disrupt trade activities and therefore there is no upper limit in these. Similarly, the **Blue box** includes a range of subsidies that aims to reduce the trade distortion and hence are without any upper limit.
- **Statement 2 is correct:** The **Amber Box** subsidies deal with a range of subsidies and are capped at 5% of the agricultural production (10 % in case of developing countries.) The concept of **peace clause** came up in the **Bali Ministerial meeting in 2013** which allowed the countries to exceed the limit subsidies for the implementation of the food security programmes.

Q.78) Ans: C

Exp:

- **Statement 1 is incorrect:** **Kessler Syndrome** deals with the situation when the density of the objects in the Low Earth Orbit is sufficient enough to cause mutual collisions with each other resulting in creation of new debris and this further enhances the chances of collision rendering the universe impassable.
- **Statement 2 is incorrect:** **Black hole** is a place where the gravitational pull is so strong that even light cannot escape. **Red star** is the first stage in the evolution of a star where the star becomes unstable after exhausting its fuel (hydrogen) and turns red. Formation of black is



the last stage in the evolution of a star. Every star turns into a Neutron star or Black Hole depending on its mass as given by **S. Chandrashekhar**. He proposed that any stable white dwarf star cannot be more massive than 1.44 times the mass of the sun (which is the Chandrashekhar limit). In case they acquire greater mass, they either turn into neutron stars or the black hole (if the mass is greater than 3 times that of the Sun).

- **Statement 2 is incorrect:** **Oxygen concentrators** are electrical devices that are used for the concentration of oxygen from the surrounding air. It contains materials and membranes that absorb nitrogen and produce oxygen with around 90-95 percent purity. They are light in weight portable devices that provide continuous operation once switched on. They also do not require any special arrangement for the storage of oxygen.
- **Statement 3 is correct:** **Zeolite membranes** are used for absorption of nitrogen from the air that is fed into the concentrators.
- **Statement 4 is incorrect:** **Oxygen cylinders** are dedicated containers for holding medical oxygen at high pressure. They provide high purity oxygen and are stored and transported through cryogenic tankers. They are usually made of steel, aluminium, or any other composite material and store oxygen in a non-liquid state.

Q.79) Ans: B

Exp:

- **Statement 1 is incorrect:** **Ramanujacharya** is a revered philosopher belonging to the **Vaishnavism** school of philosophy. He gave the concept of **Vishishtadvaita under the Vedanta school of philosophy**. It is also known as Qualified monism or non-dualism. As per this, a Brahman exists not in a single form but has manifestations in various forms. It believes that all diversity finally merges into the Supreme.
- **Statement 2 is correct:** There were two schools of philosophy namely the **Nirguna** and the **Sarguna** school. The **Nirguna** school regarded God to be devoid of any form or quality and believed in the attainment of knowledge. The **Sarguna** school on the other hand believed in the authority of the Veda and the presence of a Guru to guide an individual. Ramanujacharya belonged to the Sarguna school.
- **Statement 3 is correct:** He authored exposition or commentary in **Sanskrit** on the texts like **Brahma Sutra** and **Bhagwat Gita**.

Q.80) Ans: C

Exp:

- **Statement 1 is correct:** **Medical oxygen** is high purity oxygen that is used for treatments of ailments that involve a drop in the oxygen saturation levels. It comprises **95% of oxygen**. **Cryogenic distillation process** is used for the manufacturing of the **Liquid Medical Oxygen** in large power plants where compressed atmospheric air is fed into the distillation chamber.

Q.81) Ans: C

Exp:

Statement 1 is incorrect: The proposed UN Food Systems Summit of September 2021 is not the second, but the first ever to be held as highlighted by the Secretary General of the United Nations. The summit is planned to be essentially participatory and consultative and requires innovative ideas from the national, states and independent consultation.

Statement 2 is correct: The primary objective is to strategize the actions for positive change in Agri- food systems in the world to realize the vision of the 2030 Agenda for Sustainable Development. It will focus on the roadmap to shape food systems nationally and globally to accelerate progress in the Sustainable Development Goals.

Statement 3 is correct: The wide array of activities that is involved in the production, processing, transporting and consuming food is collectively known as the food system. It is directly proportional to the holistic development of an individual and the related aspects like families, communities, nations, environment along with its economic and cultural developments.



Statement 4 is incorrect: The Ministry of Agriculture and Farmer Welfare has created a website to offer opportunities to various stakeholders to involve and provide inputs/ ideas in context of the UN Food Systems Summit.

Q.82) Ans: C

Exp:

Statement 1 is incorrect: To counter China's dominance of supply chain in the Indo-Pacific region, the trade ministers of India, Japan and Australia have formally launched the Supply Chain Resilience Initiative. The above-mentioned three countries along with the United States of America forms the Quad group, so it is not launched by the Quad group.

Statement 2 is correct: Supply Chain Resilience is an approach in international trade practices where a country ensures that it has a diversified set of supply from a bunch of nations rather than dependent on one nation or few for its supply and need. It reduces the risk associated with supply mechanisms in case of any unanticipated events like the covid-19 pandemic or any natural/man made disasters.

Statement 3 is correct: The Supply Chain Resilience Initiative was first proposed by Japan and aims to reduce dependence on China and to make the Indo-Pacific region resilient in matters of supply chain to any adverse consequences like that of the covid-19 pandemic.

The Supply Chain Resilience Initiative will focus on sharing of best practices on supply chain resilience and should be reliable, dependable suppliers to each other. Events like investment promotion mechanisms, buyer-seller matching events to be conducted to explore the possibilities of diversification of supply chain for the stakeholders. It also aims to create a virtuous cycle of enhancing supply chain resilience with a view to eventually attaining strong, sustainable, balanced and inclusive growth in the Indo-Pacific region.

Q.83) Ans: D

Exp:

Statement 1 is correct: Convalescent Plasma theory is based on the old concept of developing immunity from the antibodies. It uses the antibodies from the recovered patients. Passive immunity is provided when a person is given antibodies to a disease rather than producing them through his or her own immune system.

Statement 2 is correct: In Convalescent Plasma (CP) therapy the whole blood or plasma from a recently cured person is taken and the plasma is then injected into the critically ill patients so that the already synthesized antibodies are transferred and boost their fight against the virus. By injecting the plasma at an early stage to an infected patient, it can prevent severe illness.

Statement 3 is correct: Plasma is the clear, straw-colored liquid portion of blood that remains after red blood cells, white blood cells, platelets and other cellular components are removed. It is the single largest component of human blood, comprising about 55 percent, and contains water, salts, enzymes, antibodies and other proteins.

Statement 4 is correct: This therapy has been used several times in the past, for example the US used plasma of recovered patients to treat patients of Spanish flu (1918-1920). Recently in 2014, WHO used it to treat Ebola and in 2015, CP therapy was used for treating MERS patients.

Q.84) Ans: B

Exp:

Statement 1 is incorrect: African swine fever (ASF) is different from Swine flu, and does not affect people and there is no impact on human health as it only spreads from animals to other animals. It was first detected in Africa in the first part of the 20th century.

Statement 2 is correct: It is a severe viral disease that affects the wild and domesticated pigs typically resulting in an acute haemorrhagic fever. The mortality rate is close to 100%, and since the fever has no cure, the only way to stop its spreading is by culling the animals as it can result in a severe transboundary spread disease.



Q.85) Ans: A

Exp:

The World Economic Forum has released the Global Gender Gap Report 2021.

Statement 1 is correct: India is now one of the worst performers in South Asia, trailing behind neighbours Bangladesh, Nepal, Bhutan, Sri Lanka and Myanmar. India has fallen 28 places- it is now ranked 140 among 156 countries.

Statement 2 is correct: The estimated earned income of women in India is only one-fifth of men's, which puts the country among the bottom 10 globally on this indicator.

Statement 3 is incorrect: The top 10 Gender Equal countries are as follows:

1. Iceland 2. Finland 3. Norway **4. New Zealand** 5. **Rwanda** 6. Sweden 7. **Namibia** 8. Lithuania 9. Ireland 10. Switzerland.

Statement 4 is correct: The year 2021 is the 15th edition of the Global Gender Gap Index. The Global Gender Gap Index was first introduced by the World Economic Forum in 2006 to compare countries' gender gaps across four dimensions: economic opportunities, education, health and political leadership.

Q.86) Ans: C

Exp:

Statement 1 is incorrect: IMPRINT is a national initiative of the Ministry of Education, designed to address major engineering challenges relevant to India through an inclusive and sustainable mode of translational research.

Statement 2 is correct: IMPRINT is a joint initiative of the Indian Institutes of Technology and the Indian Institutes of Science.

Statement 3 is correct: It seeks to develop a road for research to solve major engineering and technology challenges in environment and climate along with other domains like

- Health care
- Information and communication technology
- Energy
- Sustainable habitat,
- Nano-technology hardware
- Water resources and river systems.
- Advanced materials.
- Manufacturing.

- Security and defence.
- Environment and climate.

Q.87) Ans: D

Exp:

The Union Ministry of Environment, Forest and Climate Change has proposed several amendments to the Forest (Conservation) Act, 1980 (FCA), which may enable infrastructure projects to come up in the forest areas more easily. The proposal also aims to empower state governments to lease forest land to private individuals and corporations. If the proposed amendments come into force, they would dilute the provisions of the landmark 1996 decision of the Supreme Court in Godavarman case (TN Godavarman Thirumulpad vs Union Of India & Ors).

The proposed amendments are as follows:

- To complete the process of forest identification in a time-bound manner
- To enable the creation of 'no-go' areas, where specific projects would not be allowed.
- Proposal to grant exemptions to railways, roads, tree plantations, oil explorations, wildlife tourism and strategic projects in forests.
- They Seek to empower state governments to lease forest land to private individuals and corporations.

Forest (Conservation) Act, 1980

The Forest (Conservation) Act, 1980 an Act of the Parliament of India to provide for the conservation of forests and for matters connected therewith or ancillary or incidental thereto. It was further amended in 1988. This law extends to the whole of India. It was enacted by Parliament of India to control further deforestation of Forest Areas in India. The act came into force on 25 October 1980. It has five sections. **The proposed amendments seek to make additions and changes in Section 1 and Section 2.**

Q.88) Ans: D

Exp:

Stand up India Scheme was launched on 5th April 2016 to promote entrepreneurship at grassroot level focusing on economic empowerment and job creation. This scheme has been extended up to the year 2025.



The objective of Stand-Up India is to promote entrepreneurship amongst women, Scheduled Castes (SC) & Scheduled Tribes (ST) categories, to help them in starting a greenfield enterprise in trading, manufacturing and services sector, by both ready and trainee borrowers.

The purpose of Stand-Up India is to:

- promote entrepreneurship amongst women, SC & ST category.
- Provide loans for setting up greenfield enterprises in manufacturing, services or the trading sector & activities allied to agriculture by both ready and trainee borrowers
- facilitate bank loans between Rs.10 lakh to Rs.1 crore to at least one Scheduled Caste/Scheduled Tribe borrower and at least one-woman borrower per bank branch of Scheduled Commercial Banks.

There are certain eligibility criteria that need to be fulfilled by the people applying for the loan:

1. The individual must be **18 years or above**
2. The company must be a private limited/LLP or a partnership firm with **51% stake and control in the hand of a SC/ST or women entrepreneur.**
3. The entrepreneur should either be a woman for a person belonging to scheduled caste or scheduled tribe category.
4. The **loan will only be provided to fund greenfield projects** i.e., the project must be a very first one being undertaken under the manufacturing or service sector.
5. The applicant must **not be a bank or any other Organisation's defaulter.**
6. The company should be dealing with any commercial or innovative consumer goods. An approval of DIPP is also required for the same

Q.89) Ans: C

Exp:

Researchers at the Indian Institute of Technology, Delhi have developed a handheld Surface Enhanced Raman Spectroscopy (SERS)-based platform for early diagnosis of dengue and also gives dengue test results within one hour (rapid diagnosis).

The institute said that the handheld device has been successfully tested on the clinical blood samples collected from hundreds of individuals in collaboration with ICMR-National Institute of Malaria Research (NIMR), New Delhi, and also helps in rapid detection of HIV. On the other hand the conventional RT PCR technique takes much longer time.

Early detection is an important part of getting cured from it, considering India's huge population with limited medical infrastructure.

The project was funded by the IMPRINT India programme of the Ministry of Education.

Q.90) Ans: A

Exp:

Russian Scientists have launched one of the world's biggest underwater neutrino telescopes called Baikal GVD (Gigaton Volume Detector) in the waters of Lake Baikal, the world's deepest lake situated in Siberia.

It is one of the three largest neutrino detectors in the world along with the IceCube at the South Pole and Antares in the Mediterranean Sea.

To study the neutrinos, is to possibly determine the source of these elusive fundamental particles and give a light to the theory of making of the planet earth. It will help in the understanding of the origins of the universe as they were present during the Big Bang Explosion while others continue to be formed as a result of supernova explosions or because of nuclear reactions in the sun.

Neutrinos are one of the fundamental particles which make up the universe. They are also one of the least understood. A neutrino is a subatomic particle that is very similar to an electron, but has no electrical charge and a very small mass, which might even be zero. Neutrinos are one of the most abundant particles in the universe. Because they have very little interaction with matter, however, they are incredibly difficult to detect.

Lake Baikal is a rift lake located in Southern Siberia, Russia, between Irkutsk Oblast to the northwest and Buryat Republic to the southeast. It is the world's deepest lake and the seventh largest lake in the world

by surface area.

Q.91) Ans: b

Exp:

- **Statement 1 is Incorrect:** European Union (EU) has brought a law to establish strong Right to Repair laws to promote circular economy in the EU.
- **Statement 2 is Correct:** Under Right to repair, manufacturers of appliances will have to ensure spare parts are available for up to a decade (10 years). New devices will also have to come with repair manuals and be made in such a way that they can be dismantled using conventional tools when they really can't be fixed anymore, to improve recycling.

Q.92) Ans: d

Exp:

- **Statement 1 is Correct:** IUCN has categorised, savanna elephant as "endangered" and much smaller, lighter forest elephant as "critically endangered". These two are subspecies of African elephants. Both species are facing threat from poaching for ivory and human encroachment.

Species	Old Status	New Status
Forest Elephant (Loxodonta cyclotis)	Vulnerable VU	Critically endangered CR
Savanna/ Bush elephant (Loxodonta africana)	Vulnerable VU	Endangered EN
Asian Elephant	Endangered EN	

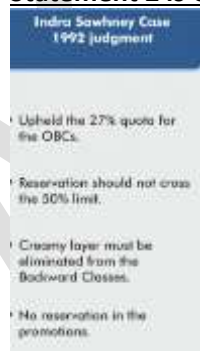
- **Statement 2 is Correct :** Asian elephants (found in India) are smaller than African elephants, and their ears are smaller compared to the large fan-shaped ears of the African species. Only some male Asian elephants have tusks, while both male and female African elephants grow tusks. Asian Elephants have 3 subspecies: Indian, Sri Lankan and Sumatran.
- **Statement 3 is Correct :** Elephants have the longest gestation period of all mammals, up to 22 months

Q.93) Ans: b

Exp:

- **Statement 1 is Incorrect :** Haryana will become 2nd State after Andhra Pradesh to have 75% reservation for locals based on Domicile (under the Andhra Pradesh Employment of Local Candidates in the Industries / Factories Act, 2019).
 - Many other states like Karnataka (100%), Maharashtra (80%) and Madhya Pradesh (70%) have also proposed such reservations for locals in the last few years.

- **Statement 2 is Correct**



- **Statement 3 is Correct :** The Constitution of India guarantees freedom of movement and consequently employment within India through several provisions.
 - Article 14 provides for equality before law irrespective of place of birth.
 - Article 15 guards against discrimination based on place of birth.
 - Article 16 guarantees no birthplace-based discrimination in public employment.
 - Article 19 ensures that citizens can move freely throughout the territory of India.

Q.94) Ans: b

Exp:

- After 170 years, a 'lost' bird was found in the jungles of Indonesian Borneo. Its Scientific Name is *Malacocincla perspicillata*. It is a songbird species in the family Pellorneidae. It is a small brown-grey songbird with a distinctive broad, black eye-stripe running all the way around its head. The species is endemic to Borneo.



- **Statement 1 is Incorrect** : The island of Borneo is governed by three separate sovereign political powers. The Island boasts of Three Nations -:
 - Brunei : A country in itself
 - Pulau Borneo: Part of Malaysia
 - Kalimantan: Part of Indonesia: It is new Capital of Indonesia
- **Statement 2 is Correct** : It is Third-largest island in the world and the largest in Asia
- **Statement 3 is Incorrect** : Borneo Island is situated at the geographic centre of Maritime Southeast Asia, in relation to major Indonesian islands, it is located north of Java, west of Sulawesi, and east of Sumatra.

Q.95) Ans: d

Exp:

- **Statement 1 is Correct** : The Sugamya Bharat App is a simple to use Mobile App with an easy registration process, requiring only 3 mandatory fields, namely, Name, Mobile number and Email-id. The App is made accessible for ease of use for persons with disabilities also with features such as font size adjustment, color contrasting option, text to speech, and having an integrated screen reader in Hindi and English.
- **Statement 2 is Correct** : The app will enable people with disabilities and even the elderly to register accessibility-related problems in buildings, in modes of transport or any infrastructure by uploading pictures on it. They can also receive information on accessibility related issues through the app. The App has the provision of easy photo uploads with a geotagging option of the premise where accessibility intervention is required.
- **Statement 3 is Correct** : Sugamya Bharat Abhiyan is conceptualised by the Department of Empowerment of Persons with Disabilities (DEPWD) as a nation-wide flagship campaign for achieving universal accessibility. The initiative is in line with Article 9 of the (UN Convention on the Rights of Persons with Disabilities) which India is a signatory since 2007. The scheme also comes under the Persons with Disabilities Act, 1995 for equal Opportunities and protection of rights which

provides non-discrimination in Transport to Persons with Disabilities.

Q.96) Ans: b

Exp:

- **Statement 1 is Incorrect** : CIIs, as defined by the Information Technology Act of 2000, are vital computer resources that, if incapacitated or destroyed, will leave a debilitating impact on national security, economy, public health or safety across both public and private sectors.
 - CI organizations in multiple sectors such as power and utilities, telecommunications and transport, and strategic and public enterprises operate according to the guidelines issued by NCPIIC.
 - These guidelines cover entire cyber security lifecycle including planning, implementation, operations, disaster recovery, and business continuity planning as well as reporting and accountability.
- **Statement 2 is Correct** : In 2014, India established the National Critical Information Infrastructure Protection Centre (NCIIPC) to regulate and protect the nation's Critical Information Infrastructures (CIIs).
Other CIIs protection in India -:
 - Indian Computer Emergency Response Team (CERT-In): National nodal agency for responding to computer security incidents.
 - National Cyber Security Coordinator (NCSC) under National Security Council Secretariat coordinates with different agencies at the national level for cyber security matters.
 - National Cyber Coordination Centre to generate necessary situational awareness of cyber security threats and enable timely information sharing for proactive, preventive and protective actions.

Q.97) Ans: d

Exp:

- **Statement 1 is Correct**: Recently, the UN Department of Economic and Social Affairs

released State of the world's indigenous peoples: Rights to Land, territories and resources report. It mentions the recognition and protection of land titles and tenure of indigenous people is crucial to attaining SDGs to eradicate poverty and achieve sustainable development by 2030. Report calls on the UN to include indigenous peoples and their organizations in the development of sustainable development frameworks.

- **Statement 2 is Correct:** Indigenous peoples are inheritors and practitioners of unique cultures and ways of relating to people and the environment. They have retained social, cultural, economic and political characteristics that are distinct from those of the dominant societies in which they live. Indigenous peoples in India comprise an estimated population of 104 million or 8.6% of the national population, almost 90% of them living in rural areas.
- **Statement 3 is Correct:** Indigenous and Tribal Peoples Convention 1957 was the First international treaty to specifically deal with indigenous rights, concerning the Protection and Integration of Indigenous and other Tribal and Semi-Tribal Populations in Independent Countries.
 - Article 1 indicates that self-identification as indigenous or tribal shall be regarded as a fundamental criterion for determining the groups to which the provisions of this Convention apply. It was replaced with Indigenous and Tribal Peoples Convention, 1989

- **Statement 2 is Incorrect:** Effect on health by different heavy metals
 - **Nitrates:** Blue baby syndrome (methemoglobinemia) which reduces oxygen level in blood
 - **Arsenic:** Black foot disease, cancer and skin lesions

Q.99) Ans: b

Exp:

- **Statement b is Correct**
- Two new species of seaweed have been discovered by a group of marine biologists from Central University of Punjab, Bathinda. They are Named *Hypnea indica* (after India) and *Hypnea bullata* (because of the blister like marks on its body – bullate), the seaweeds are part of the genus *Hypnea* or red seaweeds
- ***Hypnea indica*** was discovered Kanyakumari in Tamil Nadu, and Somnath Pathan and Sivrajpur in Gujarat
- ***Hypnea bullata*** was discovered from Kanyakumari and Diu island of Daman and Diu.
- They grow in the intertidal regions of the coast, namely the area that is submerged during the high tide and exposed during low tides.

Q.100) Ans: b

Exp:

- **Recently,** The Prime Minister has called upon to explore the possibility of e-marketing of Channapatna toys across the country and the world, during the India Toy Fair 2021.
- **Channapatna Toys :** Channapatna toys are a particular form of wooden toys (and dolls). Artisans of the town **Channapatna in Karnataka** manufacture them
 - In Kannada, Channapatna is also called "Gombegala Ooru", which means toy town in English.
 - The origin of these toys can be traced to the reign of Tipu Sultan. He invited artisans from Persia to train the local artisans in the making of wooden toys
 - The Father of the Channapatna toy is Bavas Miyan. He is known for his commitment to helping the local artisans with the art of toy-making and

Q.98) Ans: a

Exp:

- **Statement 1 is Correct:** Heavy metal refers to any metallic chemical element that has a relatively high density and is toxic or poisonous at low concentrations. Examples include mercury (Hg), cadmium (Cd), arsenic (As).
 - Density of more than 5 g/cm³ is a Commonly used criterion.
 - It is found that two-thirds of water Quality stations on major rivers have presence of one or more heavy metal beyond limits specified by the Bureau of Indian Standards.

bringing in new technologies to improvise the toys.

- Channapatna toys are made of specific wood, called "Aale Mara or the ivory wood". Vegetable dyes are used for colors that are safe to use and environment-friendly.
- This traditional craft is protected as a geographical indication(GI) under the World Trade Organization(WTO)

- **Traditional toys of Different States**

- **Kerala-** Kathakali dance dolls, animal-shaped toys, and Woodcraft Toys
- **Tamil Nadu-** Thanjavur Thalaiyatti Bommai , Choppu saman Toys
- **Punjab-** Folk toys, Handwai, Chankana, Lattu, Ghuggu
- **Telangana:** Nirmal Toys
- **Madhya Pradesh-** Adivasi Gudia Hastashilpa (Dolls), Betel Nut's Toys, and Tin Toys.
- **Uttar Pradesh-** lacQuered toys and miniature utensils Toys.
- **Jammu & Kashmir-** Walnut wood carving toys
- **Chhattisgarh-** Clay & Terracotta, Dhokra Metal Casting, Wood Carving, Metal Craft toys.
- **Daman & Diu-** Tortoise Shell toys and crafts,
- **Andaman & Nicobar Island** - Timber Toys, Coconut shell toys

- **Note -:** This Q.uestion can be solved by elimination. It doesn't reQ.uire remembering all traditional toys' names and states. Channapatna was in news and Thanjavur is a well known place in art and culture which is in Tamilnadu. Hence elimination of pair 1 and 5 gives the correct answer.



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